

INTELLICASH

VSLA Digitisation Platform

System Design Document

VERSION 1.0 · MASTER AUTHORISED RELEASE

The definitive technical specification for the design, build, and deployment of Intellicash

Developed & Owned by: Intelli-Wealth Limited (IWL)
FLOURISH VSLA Programme · Agreement SF 00112775
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This document is the single authorised source of truth for Intellicash system design. All development, testing, partner integration, and deployment decisions must reference this document. No feature may be built that is absent from this specification, and no feature specified herein may be omitted from the production system, without a formal version amendment approved by IWL Executive Director.

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SECTION 1

Vision, Mission & Strategic Context

Why Intellicash exists and what it must achieve

1.1 Platform Vision

To be the most trusted digital infrastructure for grassroots savings and credit in Africa — making every shilling saved by every VSLA member fully transparent, secure, and connected to the formal financial system.

1.2 Platform Mission

Intellicash exists to digitise the complete lifecycle of a Village Savings & Loan Association (VSLA) — replacing the physical cashbox, paper passbooks, and verbal group votes with a secure, transparent, offline-capable digital system that preserves and enforces the IWL Savings Group Standard (IWLSGS) methodology at every step.

The platform must simultaneously serve:

- VSLA members — enabling transparent personal financial records, dignified loan access, and democratic participation
- Group officers and key-holders — enforcing fiduciary duty, procedural integrity, and collective governance
- Village Agents / CBTs — making field facilitation measurable, auditable, and scalable
- Partner organisations (NGOs, MFIs, banks, government) — providing real-time portfolio intelligence without compromising group sovereignty
- Intelli-Wealth Limited — advancing financial inclusion, generating programme evidence, and building a sustainable social enterprise

1.3 Strategic Context

The VSLA model is one of the most proven financial inclusion tools in Sub-Saharan Africa, with over 14 million members across the continent. Yet the model remains largely paper-based, vulnerable to fraud, invisible to formal finance, and difficult to monitor at scale. Intellicash is IWL's answer to these structural limitations.

Strategic Challenge	Intellicash Response
Physical cashbox risk: single-point fraud by treasurer or officer	Three-Key OTP system: no meeting opens without all three key-holders physically present and digitally verified
Paper passbooks: easily falsified, lost, or manipulated	Append-only blockchain-backed digital passbook: cryptographically immutable, permanently auditable

Strategic Challenge	Intellicash Response
Groups invisible to formal finance: no credit history	IWL Credit Score: machine-computed from live group data; credentialing groups for bank and MFI access
Groups without bank accounts excluded from formal credit	IWL Custodian Paybill: groups save into IWL-managed sub-accounts via M-Pesa, enabling formal-equivalent credit access
Members unable to guarantee each other's loans formally	Credit Guarantee System: blockchain-recorded guarantor commitments with automatic liability calling on default
Donors and partners flying blind on programme impact	Real-time MEL dashboard and API: live data on 47 indicators across all programme groups
VA field work unaccountable and difficult to quality-assure	CBT Module: GPS-stamped structured visit logs, quality scoring, phase gating — all on a single mobile app
External loans inaccessible to most VSLAs	External Loan Module: direct integration with KCB, Equity, and AMREF MFI; IWL Direct Loans via Paystack

1.4 Design Principles

These principles are non-negotiable. Any feature that conflicts with a design principle must be redesigned, not the principle relaxed.

Principle	Statement	Implication for Developers
Methodology-First	Every screen, workflow, and data field must map to a documented IWLSGS procedure	No feature may be added that bypasses or contradicts IWL's VSLA methodology
Offline-First	Full meeting functionality — all 8 steps — must work without internet connectivity	All financial logic must run on-device; server sync is secondary, not blocking
Blockchain-Immutable	Every financial transaction must produce an immutable blockchain record	No DELETE or UPDATE ever permitted on any financial record in any database layer
Group Sovereignty	The platform cannot override a General Assembly decision	No admin, officer, or system function can reverse a GA-approved vote result
Three-Key Integrity	No meeting can open, and no money can move, without all three key-holders present and verified	OTP activation is a hard gate — not a soft warning — at every meeting entry point
Low-Literacy UX	Every screen must be usable by a member with no formal education	Icon-first design; audio prompts; Kiswahili primary; no complex text without visual support

Principle	Statement	Implication for Developers
Tri-Account Clarity	Every user account type (Member, Group, Agent) must have a distinct, non-overlapping UI experience	Account type is set at registration and cannot be changed without Admin approval
Financial Pass-Through	Intellicash never holds member funds as a financial institution	All funds flow through M-Pesa, Paystack, KCB Buni, or IWL Custodian — never resting in the platform itself
Privacy-by-Design	Member PII must never appear on the blockchain; KDPA 2019 compliance is non-negotiable	All PII encrypted at rest; only hashes on-chain; right to erasure via anonymisation
API-Open	All data must be accessible to authorised partners via a well-documented REST API	Every major data entity must have a corresponding read-only API endpoint from Day 1

SECTION 2

User Personas & Account Types

Who uses Intellicash, their goals, and their challenges

2.1 Tri-Account Onboarding Model

Every user who installs Intellicash must choose one of three account types at first registration. This is a hard architectural boundary — the account type determines every screen, permission, and data access path the user will ever see. There is no account type switching without a full re-registration process approved by IWL Admin.

SECTION 2 — USER PERSONAS & ACCOUNT TYPES

Intellicash Tri-Account Model

Three hard-bounded account types · Each determines every screen, permission, and data access path · Immutable after first registration

MEMBER

Individual VSLA member

"Save, borrow & vote in my group"

PERSONAS

- Amina Wanjiku
Ordinary member · Own Android · 2G/3G · Low digital literacy
- Joseph Mwangi
Chairperson · Group tablet · Cannot approve own loans
- Grace Atieno
Treasurer · Manages B2C disbursements · Dual-OTP rules
- Daniel Ochieng
Key-Holder · 6-digit OTP via SMS · GPS-verified

CORE ACCESS

- Personal passbook — real-time view
- Buy 1-5 shares via M-Pesa
- Apply for internal loans
- Anonymous GA votes
- Guarantee fellow members' loans
- Receive share-out payout

IWL Green (#1B6B3A) UI Theme

GROUP

VSLA group entity — managed by elected officers

"Manage our VSLA account collectively"

PERSONA

- Unity Chama VSLA
Group-shared tablet managed by officers · Paybill accessible from any phone

CORE ACCESS

- 8-step meeting workflow management
- Collect savings via M-Pesa Paybill
- Apply for external loans (bank/MFI/IWL)
- Conduct anonymous GA votes
- Track all 4 fund sub-accounts
- Build IWL Credit Score (0-100)
- Share-Out Wizard at Action Audit

IWL Blue (#0D47A1) governance screens

AGENT

Village Agent / CBT · IWL Staff · Partner Officer

"Form, train, monitor & link groups"

PERSONAS

- Peter Kamau (VA/CBT)
15-25 groups · IWL tablet · GPS field work · 3G/4G SIM
- Margaret Waiganjo (IWL Staff)
IWL Loan Officer · Web Platform · Reviews IWL Direct Loans
- Dr. Njoroge (Partner/MFI)
AMREF Partner Officer · Web browser · Read-only portfolio access

CORE ACCESS

- CBT Module — Group Formation Wizard
- GPS-stamped structured visit logs
- 9 IWL training module delivery
- Phase advancement (requires Admin counter-sign)
- VA Network (VAN) peer meetings
- MEL dashboards & API (partners)

IWL Purple (#4A148C) Agent UI Theme

MOBILE APP — 5-TAB NAVIGATION BY ACCOUNT TYPE

TAB	MEMBER	GROUP (OFFICERS)	AGENT (CBT)
Dashboard	Personal shares, loan balance, guarantor liability, recent transactions, next meeting countdown	Group fund balances, credit score, active loans, meeting status, sync indicator	Caseload overview, overdue visits, pending phase advancements, performance
Meetings	View meetings, cast votes during session, confirm OTP for transactions	Start New Meeting, 3-key OTP activation, 8-step workflow, minutes recorder	Not primary tab — linked via CBT Module check-in gate
Members	View own profile, KYC status badge, passbook history	Full roster with search/filter, role assignment, member KYC management, credit scores	Group member roster for each caseload group
Main	Loan book: applications, repayment schedule, guarantor view	Full loan book with disbursement controls, guarantors, external finance	CBT: caseload dashboard, visit log, training modules, phase gate, VAN
More	Transactions, settings, account management, help & audio guides, sync status	Group Finance, Share-Out Wizard, Reports, settings, Paybill deposits	Group Formation Wizard, training modules, VAN network, performance reports

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2.2 Member Account Personas

□ Amina Wanjiku

Ordinary VSLA Member · Account Type: MEMBER

Device / Access	Own Android smartphone (low-to-mid range, 2G/3G)
Primary Goals	• View and track her personal digital passbook at any time • Buy 1–5 shares per meeting using M-Pesa • Apply for and receive internal loans • Cast anonymous votes in General Assembly • Guarantee a fellow member's loan • Receive share-out payout at end of cycle
Key Challenges	• Low digital literacy — needs icon-first UI with Kiswahili labels • Fear of digital fraud — prefers to see transactions confirmed publicly • Intermittent mobile data and power access • Does not trust systems she cannot explain to her neighbours

□ Joseph Mwangi

Group Chairperson · Account Type: MEMBER (Officer)

Device / Access	Group-owned shared tablet (Android, managed by IWL)
Primary Goals	• Open and close meeting sessions (requires 3-key OTP to be verified) • Table resolutions and oversee the 8-step meeting agenda • Cannot approve his own loan application or vote on his own motion • Oversee compliance with VSLA constitution • Escalate disputes and anomalies to the VA or IWL Admin
Key Challenges	• Pressure from members to speed up or skip steps — platform must enforce order, not rely on him • Risk of over-reach — system must enforce separation of duties • Managing group dynamics with a digital tool that some members distrust

□ Grace Atieno

Group Treasurer · Account Type: MEMBER (Officer)

Device / Access	Group-owned shared tablet
Primary Goals	• Confirm fund counts after each step — Loan Fund, Social Fund, External Fund • Initiate and confirm M-Pesa B2C disbursements within active sessions • Run the Share-Out Wizard at Action Audit • Reconcile paybill deposits from IWL Custodian Account • Cannot disburse to herself without dual officer OTP

Key Challenges

- M-Pesa delays mid-meeting create member anxiety
- Reconciling offline records with sync'd data after connectivity outage
- Dual-authority rules slow operations — platform must make dual sign-off fast, not bureaucratic

□ Daniel Ochieng

Key-Holder (1 of 3) · Account Type: MEMBER (Key-Holder)

Device / Access

Own phone — must be present at meetings

Primary Goals

- Receive 6-digit OTP via SMS at start of each meeting
- Read out OTP aloud to the group; Secretary enters it into the Group Account
- GPS must confirm Daniel is within 50m of meeting location
- Cannot simultaneously hold a Management Committee officer role

Key Challenges

- SMS delays in areas with weak Safaricom signal
- Phone battery — must arrive with charged phone
- Social pressure if he is late and blocking the meeting from starting

2.3 Group Account Personas

□ Unity Chama VSLA

VSLA Group Entity (managed by elected officers) · Account Type: GROUP

Device / Access

Group-shared tablet managed by officers; Paybill accessible from any phone

Primary Goals

- Manage group meetings through the 8-step workflow
- Collect savings via M-Pesa Paybill into IWL Custodian Account (no bank account)
- Apply for and manage external loans from banks, MFIs, and IWL Direct
- Conduct anonymous General Assembly votes
- Track all fund sub-accounts: Loan Fund, Social Fund, External Fund, Custodian
- Build IWL Credit Score to access formal financial services

Key Challenges

- Group has no bank account — relies on Paybill custodian model
- Managing multiple concurrent fund types without commingling
- Collective trust in the platform — if one member distrusts, others may follow

2.4 Agent Account Personas

□ Peter Kamau

Village Agent / Community-Based Trainer (CBT) · Account Type: AGENT

Device / Access

IWL-issued Android tablet; 3G/4G SIM

Primary Goals	• Form new VSLA groups through the digital Group Formation Wizard • Log GPS-stamped structured visit notes for each group in caseload • Deliver IWL's 9 training modules via the CBT Module • Score group quality using 12 IWL indicators • Advance groups through lifecycle phases (requires IWL Admin counter-sign) • Manage VAN — Village Agent Network — peer meetings and cross-group learning
Key Challenges	• Large caseloads with 15–25 groups across remote areas • Intermittent connectivity in field locations • Balancing data quality with speed during village visits

□ Margaret Waiganjo

IWL Loan Officer · Account Type: AGENT (IWL Staff)

Device / Access	Web platform + Mobile (IWL office device)
Primary Goals	• Review IWL Direct Loan applications submitted by groups • Access group credit score breakdown and passbook history • Approve and initiate Paystack disbursements to qualifying groups • Monitor repayment schedules and flag arrears • Generate loan statements for partner and regulatory reporting
Key Challenges	• Verifying group eligibility across multiple data sources • Paystack disbursement errors to unregistered phone numbers • Coordinating with groups who cannot access digital repayment channels

□ Dr. Njoroge — AMREF

Partner Officer / Bank/MFI Representative · Account Type: AGENT (Partner)

Device / Access	Web browser (office or field laptop)
Primary Goals	• View credit-eligible group portfolio via Web Platform • Access IWL Credit Score and credit data package per group • Submit group loan approval or counter-offer • Monitor external loan repayment schedules • Export MEL indicators and programme impact data
Key Challenges	• Data reliability concerns — needs immutable audit trail to trust the data • Regulatory requirements for collateral and KYC documentation • Approval workflow delays within partner organisation

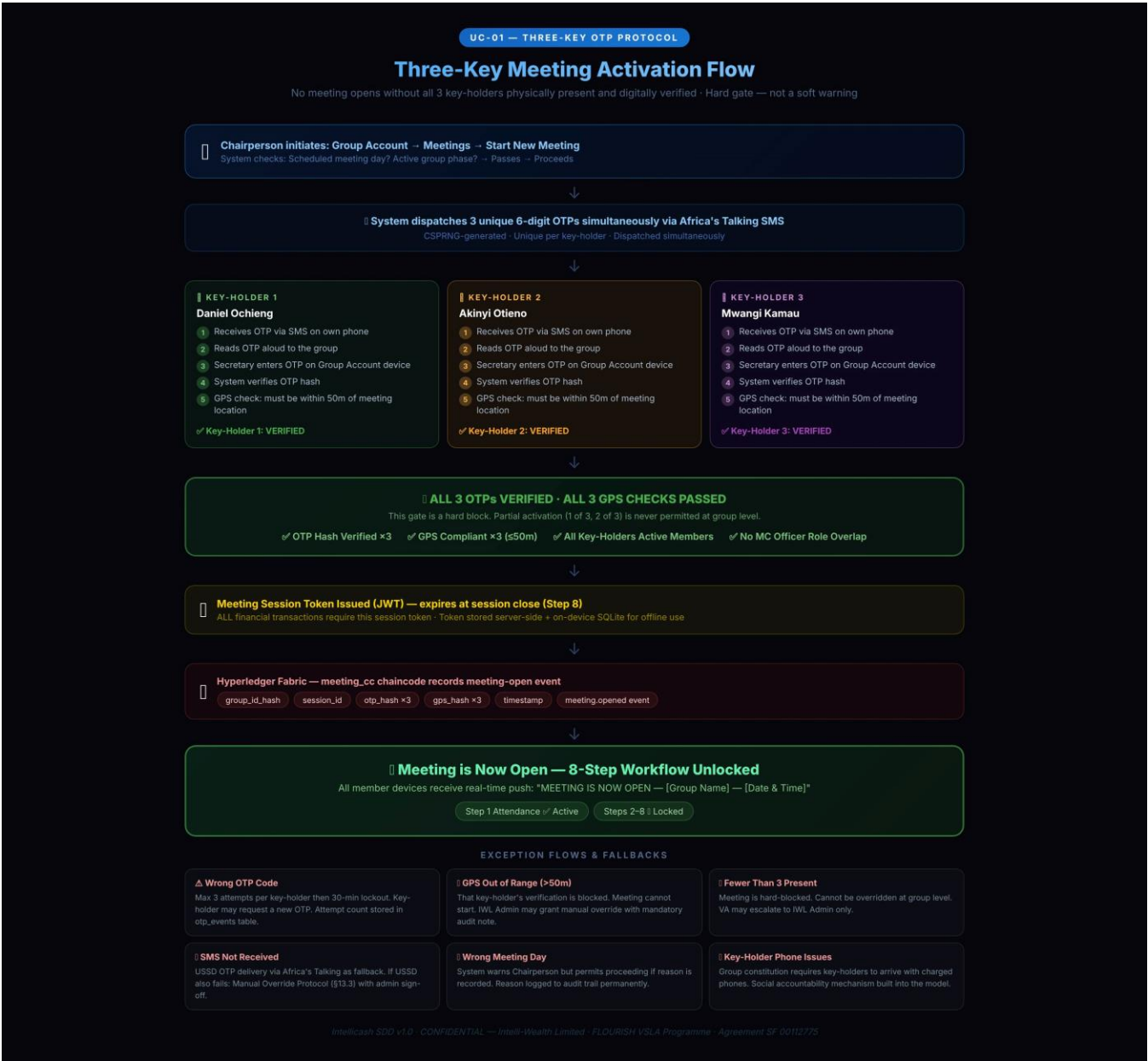
SECTION 3

Use Cases

Detailed interaction specifications for key system functions

Use cases describe the primary interactions between personas and the Intellicash system. Each use case is a binding specification for the developer — the system must implement every step exactly as described. Exception flows are not optional edge-case handling; they are required production behaviour.

UC-01: OTP-Activated Meeting — Three-Key Protocol



UC-01	OTP-Activated Meeting — Three-Key Protocol
Primary Actor	Key-Holder ×3 (Daniel, two others) + Group Chairperson + Group Secretary
Trigger	Chairperson initiates a new meeting session in the Group Account → Meetings → Start New Meeting
Pre-conditions	Group is active; 3 Key-Holders are designated and enrolled with verified phone numbers; GPS meeting location is registered; all 3 Key-Holders are physically present; meeting date matches scheduled meeting day in group constitution.
Main Flow	1. Chairperson opens Intellicash Group Account → Meetings → Start New Meeting. 2. System checks: Is today a scheduled meeting day? Is the group in an active phase? Passes — proceeds. 3. System generates three unique 6-digit OTPs via CSPRNG and dispatches via Africa's Talking SMS to each Key-Holder's registered number simultaneously. 4. Key-Holder 1 receives SMS. Reads OTP aloud to the group. Secretary enters it into the 'Key-Holder 1 OTP' field on the Group Account shared device. 5. System verifies OTP hash, checks GPS of Key-Holder 1's device (must be within 50m of registered meeting GPS). Marks Key-Holder 1: VERIFIED ✓. 6. Key-Holder 2 repeats the same process. System marks Key-Holder 2: VERIFIED ✓. 7. Key-Holder 3 repeats the same process. System marks Key-Holder 3: VERIFIED ✓. 8. All 3 OTPs verified and all 3 GPS checks passed. System issues a Meeting Session Token (JWT, expires at session close). 9. Blockchain (Hyperledger Fabric — meeting_cc chaincode) records: meeting-open event with group_id, session_id, OTP hash for each key-holder, GPS hash, timestamp. 10. All member devices connected to the group receive a real-time push: 'MEETING IS NOW OPEN — [Group Name] — [Date & Time].' 11. Meeting Step 1 (Opening & Attendance) is now active. 8-step workflow engine is unlocked.
Post-conditions	Meeting session is active with a valid session token; all financial transactions are permitted; blockchain records the open event; all member passbooks are live.
Exceptions / Alt Flows	• OTP fails (wrong code): system allows max 3 attempts per key-holder then 30-min lockout. Key-Holder may request a new OTP. • GPS out of range (>50m): system blocks that key-holder's verification; notifies Chairperson. Meeting cannot start until all 3 GPS checks pass. IWL Admin can grant a manual override with a mandatory audit note. • Fewer than 3 Key-Holders physically present: meeting is blocked. System cannot be overridden at group level. VA may escalate to IWL Admin. • OTP SMS not received (no signal): Key-Holder requests USSD OTP delivery as fallback. If USSD also fails: Manual Override Protocol (see §13.3). • Wrong meeting day: system warns Chairperson but permits proceeding if Chairperson records reason (logged to audit trail).

UC-02: Member Internal Loan Application with Credit Guarantee

UC-02	Member Internal Loan Application with Credit Guarantee
Primary Actor	VSLA Member (borrower) + Guarantor Members (1–3) + General Assembly + Treasurer
Trigger	Meeting is active (session token valid); Step 6 — Loan Applications — is current step
Pre-conditions	Meeting session is open; borrower is an active member; Loan Fund has sufficient balance; step 6 is current step in workflow.
Main Flow	1. Member states loan purpose aloud to group. Secretary captures it in the app: Group Account → Meeting → Step 6 → New Loan Application. 2. System performs automatic eligibility check: borrower's share_value × 3 = maximum loan amount. Checks displayed to group. 3. Secretary records requested amount and purpose. System auto-calculates service charge (flat % × months) and repayment schedule. 4. Member nominates 1–3 fellow active members as Credit Guarantors. System sends in-app notification to each nominated guarantor's Member Account. 5. Each nominated guarantor must Accept or Decline within the active meeting session. Liability amount is displayed clearly: 'If [borrower] defaults, you are liable for KES [amount].' 6. At least 1 guarantor accepts. All acceptances are recorded; blockchain writes guarantee.accepted event for each guarantor. 7. Chairperson tables the loan for GA vote. Anonymous ballot is opened: members tap Yes / No / Abstain on their Member Account devices. 8. Vote closes after all present members have voted or 3-minute timer expires. Quorum check: 75% Yes required. 9. Vote result is blockchain-hashed (SHA-256 of resolution_id + yes + no + abstain + timestamp). Resolution sealed. 10. If 75% Yes: system unlocks M-Pesa B2C disbursement. Treasurer initiates. Borrower's Member Account displays OTP request. 11. Borrower confirms receipt via OTP. M-Pesa B2C payment confirmed. Passbook updated live with loan entry and repayment schedule. 12. Blockchain records: loan_disbursed event with loan_id, borrower_id, amount, guarantors[], vote_hash, mpesa_ref, timestamp.
Post-conditions	Loan disbursed; passbook updated; repayment schedule active; guarantor passbooks flagged 'Guarantor Liability: KES [amount]'; blockchain record immutable; lender API updated.
Exceptions / Alt Flows	• No guarantor accepts within session: loan application is suspended. Member may resubmit at next meeting with new guarantors. • Vote fails quorum: loan rejected; Chairperson records reason; member may reapply next meeting. Blockchain records failed vote. • Loan Fund insufficient: system blocks disbursement. Chairperson may reduce loan amount or defer. • M-Pesa B2C fails: payment queued; Treasurer retries or initiates fallback. Loan status = 'Disbursement Pending' until confirmed. • Guarantor later requests to withdraw: only permitted before disbursement. After disbursement, guarantee cannot be withdrawn. • Borrower defaults: Guarantee Service calls guarantee on defaulting loan. Guarantor notified via SMS and in-app. Guarantor's share balance is flagged.

UC-03: Group Saves via IWL Paybill (No Bank Account)

UC-03	Group Saves via IWL Paybill (No Bank Account)
Primary Actor	Group Member or Treasurer (any phone) → IWL Paybill Service → Group Treasurer (allocation)
Trigger	Group needs to deposit savings, repay an external loan, or transfer funds; group has no bank account
Pre-conditions	Group Account is active; IWL Paybill number (522522) is configured; group has a unique Account Reference (GRP-XXXXX); Treasurer is enrolled.
Main Flow	1. Member or Treasurer dials M-Pesa on any phone. Goes to: M-Pesa → Lipa Na M-Pesa → Pay Bill. 2. Enters IWL Paybill Business Number: 522522. 3. Enters group's unique IWL Account Reference as the Account Number (e.g., GRP-00445). 4. Enters amount and M-Pesa PIN. Confirms payment. M-Pesa sends confirmation SMS to sender. 5. M-Pesa Daraja v3 fires a C2B webhook to IWL API Gateway with: TransactionID, amount, AccountReference, sender phone, timestamp. 6. Paybill Service receives webhook; validates signature; matches AccountReference to group_id in group_accounts table. 7. Paybill Service creates a paybill_transactions record. Credits group's custodian_balance_kes in group_accounts. 8. Blockchain (payment_cc) records: paybill.received event with paybill_txn_id, group_id hash, amount, mpesa_txn_id, timestamp. 9. Notification Service sends SMS to Treasurer's registered phone and in-app push to Group Account: 'KES [amount] received. Reference: [GRP-XXXXX]. Please allocate to a fund.' 10. Treasurer opens Group Account → Finance → Paybill Deposits → Allocate. Selects fund: Loan Fund / Social Fund / External Loan Repayment. 11. Fund Service credits selected fund namespace. Paybill transaction record updated: allocation_fund set, allocated_at timestamp. 12. Group Account dashboard reflects updated fund balances immediately.
Post-conditions	Funds are credited to IWL Custodian sub-account; Treasurer has allocated to correct fund; group ledger is updated; blockchain record is immutable; partner Web Platform reflects updated balances.
Exceptions / Alt Flows	• Wrong account reference entered: funds land in 'unmatched_holding' queue. IWL Finance reviews within 24 hours and manually matches or refunds via M-Pesa reversal. • M-Pesa reversal received: Paybill Service handles reversal webhook; debits custodian balance; notifies Treasurer. If already allocated: Fund Service reverses allocation. • Treasurer does not allocate within 48h: system sends reminder. After 72h: IWL Admin is alerted. • Group later opens a bank account: IWL Finance processes an EFT of custodian balance to bank account on group request. Custodian sub-account is closed after confirmation.

UC-04: Group Applies for External Loan (Bank or MFI)

UC-04	Group Applies for External Loan (Bank or MFI)
Primary Actor	Group Chairperson + General Assembly + External Lender (via API or Web Platform)
Trigger	Group wants formal credit; Chairperson initiates application in Group Account → Group Finance → External Loans
Pre-conditions	Group Credit Score $\geq$ 75; meeting is OTP-activated; group has no other active external loan of the same type; GA has not rejected this application this cycle.
Main Flow	1. Chairperson opens Group Account → Group Finance → External Loans → Apply for External Loan. 2. System displays eligible lenders filtered by group credit score and lifecycle phase. Shows loan products, rates, and terms for each. 3. Chairperson selects lender and loan product. Enters requested amount and purpose. 4. System compiles IWL Credit Data Package: credit score, passbook summary, meeting attendance history, repayment record, constitution compliance score. 5. Chairperson tables as GA Resolution. System opens anonymous ballot. 75% Yes required. 6. Vote result is blockchain-hashed. On passing: system submits credit data package to lender API (or lender receives notification to review via Web Platform). 7. Lender reviews on Web Platform. Issues approval, counter-offer, or rejection. 8. On approval: disbursement is routed to group's External Loan Sub-Account (if bank account exists) or IWL Custodian Account (if Paybill-only group). 9. Group Account notified: 'External loan of KES [amount] approved by [Lender]. Disbursement confirmed.' 10. External Loan Sub-Account is credited. Repayment schedule loaded into Group Account → Group Finance → External Loans → Active Loans. 11. Blockchain records: extloan.disbursed event with application_id, lender_id, group_id hash, amount, vote_hash, disbursement route, timestamp.
Post-conditions	External loan credited to segregated sub-account; repayment schedule active; group ledger reflects new liability; lender API shows active loan; partner Web Platform dashboard updated.
Exceptions / Alt Flows	• Credit score below 75: system blocks application and shows factor breakdown, advising group on how to improve each factor. • Lender declines: group notified with reason. Reapplication permitted after 30 days. • Group already has active external loan of same type: system prevents second application until first is repaid or restructured. • Disbursement to IWL Custodian (no bank account): group must sign IWL Custodian Agreement before disbursement routes to Paybill. • Group defaults: Early Warning System triggers when repayment rate drops below 70%. IWL Admin and VA are alerted. Loan restructuring workflow is activated.

UC-05: IWL Direct Loan Application via Paystack

UC-05	IWL Direct Loan Application via Paystack
Primary Actor	Group Treasurer + General Assembly + IWL Loan Officer (Web Platform)
Trigger	Group applies for a direct loan from Intelli-Wealth Limited; Group Account → Group Finance → IWL Direct Loans
Pre-conditions	Group Credit Score ≥ 80 ; group has completed at least 1 full operating cycle; no active IWL Direct Loan outstanding; GA vote has not been conducted this cycle for IWL loan.
Main Flow	1. Treasurer opens Group Account → Group Finance → IWL Direct Loans → Apply. 2. System checks eligibility: credit score, cycle count, outstanding IWL loan balance. If eligible: application form is displayed. 3. Treasurer enters: requested amount, loan purpose, preferred repayment period (3–12 months). 4. System auto-computes: maximum eligible amount (based on savings pool $\times$ IWL lending formula), service charge schedule, and repayment schedule. 5. Application is tabled as GA Resolution. Anonymous ballot: 75% Yes required. Blockchain-hashed. 6. On passing: application is submitted to IWL Loan Officer queue on Web Platform. Status = 'Under Review'. 7. IWL Loan Officer reviews on Web Platform: credit score breakdown, passbook PDF, meeting history, existing liabilities. 8. IWL Loan Officer approves (or adjusts amount) and initiates Paystack transfer from IWL Lending Fund sub-account. 9. Paystack transfer confirmed. Group Account notified: 'IWL Direct Loan of KES [amount] disbursed via Paystack. Repayment begins [date].' 10. Repayment schedule is loaded into Group Account. Members can repay via: (a) Paybill using loan reference, or (b) Paystack payment link. 11. Blockchain records: iwl_loan.disbursed event with iwl_loan_id, paystack_txn_ref, group_id hash, amount, officer_id, repayment_schedule_hash.
Post-conditions	Loan disbursed; repayment schedule active; IWL Finance Web Platform shows live loan status; blockchain record immutable; credit score monitoring begins for this loan.
Exceptions / Alt Flows	• Paystack transfer fails: IWL Loan Officer retries or switches to M-Pesa B2C fallback. Loan status remains 'Disbursement Pending'. • IWL Loan Officer rejects application: group notified with written reason. Reapplication after 60 days. • Group misses repayment: system auto-alerts IWL Loan Officer at Day 7 overdue. Credit score is penalised from Day 1 overdue. • Credit score drops below 80 after disbursement: monitoring only — loan already disbursed. Flagged for enhanced monitoring. • Paystack account details change: group must update via Group Account → Settings → Payment Details. IWL Admin confirms before next disbursement.

UC-06: Village Agent Forms a New VSLA Group

UC-06	Village Agent Forms a New VSLA Group
Primary Actor	Village Agent / CBT (Agent Account)
Trigger	Agent has been assigned a community; opens Agent Account → CBT Module → New Group Formation
Pre-conditions	Agent Account is active and verified; community assessment has been completed offline; IWL Programme Admin has authorised caseload assignment for this community.
Main Flow	1. Agent opens Agent Account → CBT Module → Form New Group. 2. Agent enters: group name, GPS coordinates of regular meeting location (device GPS auto-fills), community name, county, sub-county. 3. Agent registers founding members one by one: National ID, phone number, photo. System runs IPRS KYC verification for each member automatically. 4. Agent designates 5 Management Committee officers from the verified member list: Chairperson, Secretary, Treasurer, Money Counter 1, Money Counter 2. 5. Agent designates 3 Key-Holders. System enforces: Key-Holders CANNOT be MC officers. If overlap attempted: hard block with explanation. 6. Agent configures group constitution via the Constitution Wizard: share denomination (KES), max shares per meeting, Social Fund rate (flat per meeting), loan service charge (% per month), fine schedule for absence/tardiness. 7. Agent sets meeting schedule: day of week, time, frequency (weekly in Intensive phase). 8. Agent confirms GPS meeting location on map. Group is created. System auto-generates: group_id, unique Paybill Account Reference (GRP-XXXXX), IWL Credit Score starting baseline. 9. Group Account is provisioned and linked to all registered members' Member Accounts. 10. Agent submits group to IWL Programme Admin for counter-sign. Admin reviews and activates group on Web Platform. 11. Blockchain records: group.created event with group_id, agent_id, constitution_hash, GPS hash, timestamp.
Post-conditions	VSLA group is active; all members have linked Member Accounts; Group Account is ready; Paybill reference is live; group appears on VA caseload dashboard and partner Web Platform portfolio.
Exceptions / Alt Flows	• IPRS KYC fails for a member: member registered as 'Pending Verification' — limited access until manual KYC completed within 48 hours. • Duplicate phone number detected: system flags potential duplicate member. Agent must resolve (different person sharing a phone is documented with a note). • GPS location unavailable (deep rural): Agent enters coordinates manually from offline map. IWL Admin verifies via satellite imagery on Web Platform before activating. • IWL Admin rejects group: reason provided. Agent must address issues and resubmit.

UC-07: Share-Out — End-of-Cycle Action Audit

UC-07	Share-Out — End-of-Cycle Action Audit
Primary Actor	Treasurer + Chairperson + all members (Group Account + Member Accounts)
Trigger	Group has completed 12-month operating cycle; VA has signed off on Action Audit; Chairperson opens Share-Out Wizard
Pre-conditions	Cycle is complete (12 months); VA has submitted Action Audit sign-off on CBT Module; IWL Admin has confirmed Action Audit on Web Platform; all outstanding loans are resolved or written off; all external loans are repaid.
Main Flow	1. Treasurer opens Group Account → Share-Out Wizard → Begin Action Audit. 2. System runs pre-share-out audit: checks all loans are resolved; checks all external liabilities are cleared; checks any active guarantees are discharged. 3. System computes each member's share-out entitlement: $\text{shares held} \times \text{current share value} + \text{proportional interest earned} - \text{any outstanding fines or loan liabilities}$. 4. Share-out schedule is displayed to all member devices in real time: each member can see their own entitlement. 5. Chairperson tables the share-out schedule as a GA Resolution for approval. 6. GA votes: simple majority required to approve the schedule. 7. On approval: Treasurer initiates M-Pesa B2C batch payout. Each member's phone receives payment. 8. Each member confirms receipt via OTP in their Member Account. 9. Passbook entry for each member: CYCLE CLOSED — Share-Out Received: KES [amount] — [date]. 10. Blockchain records: shareout.disbursed event for each member with shareout_id, member_id, amount, mpesa_ref, otp_confirmed. 11. Group cycle counter increments. Constitution is reset to current version. New cycle begins.
Post-conditions	All members have received their share-out; all passbooks are marked CYCLE CLOSED for the completed cycle; blockchain records are immutable; partner Web Platform shows cycle completion; new cycle is active.
Exceptions / Alt Flows	• Member's M-Pesa B2C fails (unregistered number, limit reached): funds held in Pending Payout status. Treasurer manually re-initiates within 48 hours or arranges cash payment with dual officer sign-off. • Member has outstanding loan at share-out: loan balance is deducted from share-out entitlement. If loan exceeds entitlement: deficit is written off with GA vote and documented in audit log. • Active guarantor liability at share-out: guarantor's entitlement is held in escrow until borrower's loan is resolved. • Member disputes share-out amount: Chairperson opens dispute in Group Account. IWL Admin reviews passbook audit trail. Resolution within 7 days.

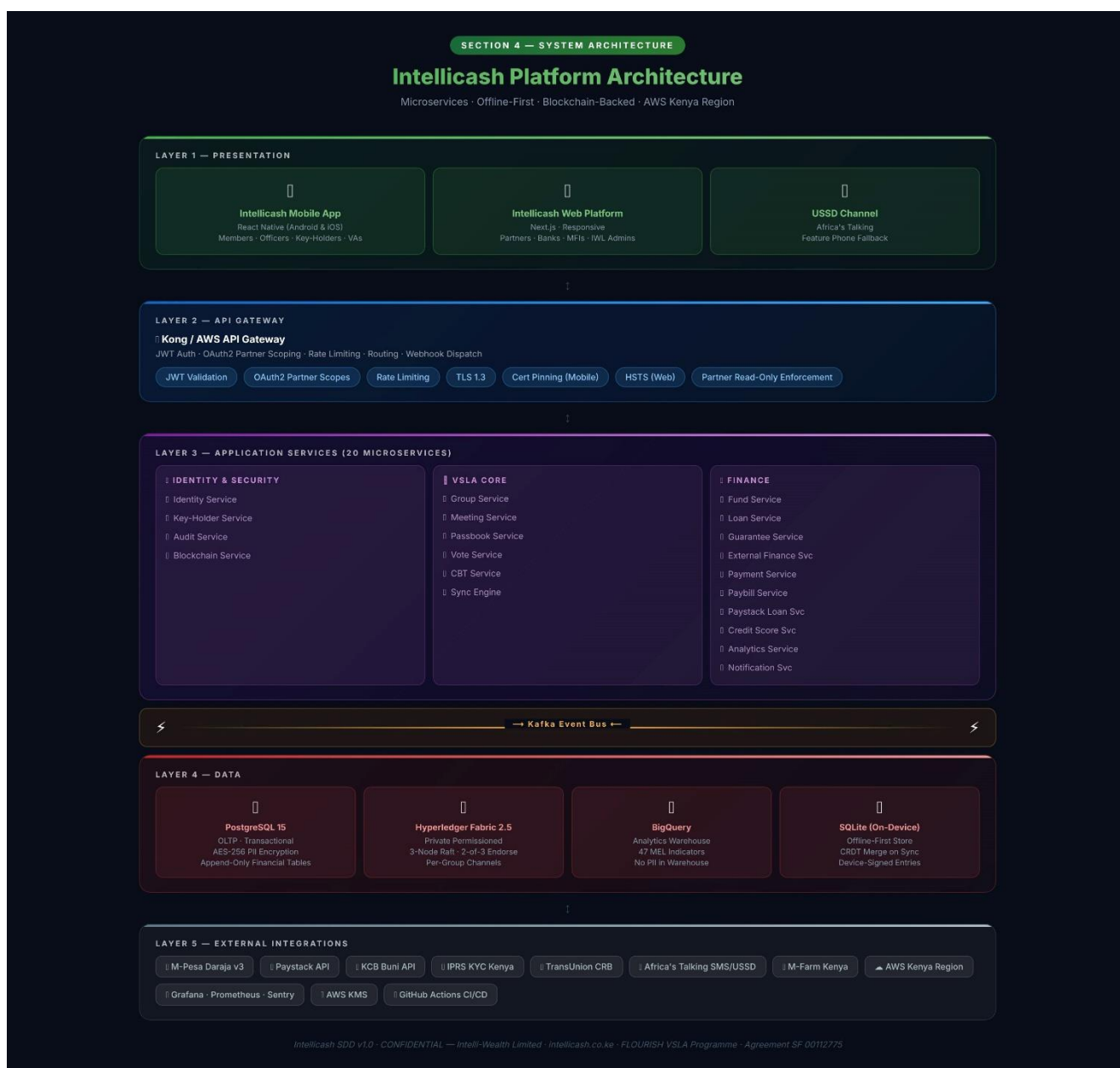
SECTION 4

System Architecture

Platform structure, layers, and technology stack

4.1 Architecture Overview

Intellicash is built on a microservices architecture with a clear separation between the mobile presentation layer (Members, Officers, Agents), the web presentation layer (Partners, Banks, IWL Admin), and the backend service layer. All services communicate through a central API Gateway. The data layer uses PostgreSQL for transactional data and Hyperledger Fabric for immutable blockchain records.



4.2 Architecture Layers

Layer	Components	Technology	Key Requirements
Presentation — Mobile	Intellicash Mobile App (Android/iOS) USSD fallback for feature phones	React Native (single codebase) Africa's Talking USSD	Offline-first; role-based UI; icon-first; Kiswahili primary; audio prompts
Presentation — Web	Intellicash Web Platform API Developer Portal	Next.js / React.js Responsive (desktop & mobile)	Read-only for partners; real-time dashboard; MEL export; API portal
API Gateway	Auth, routing, rate limiting, webhook dispatch, partner scope enforcement	Kong Gateway / AWS API Gateway	JWT validation; OAuth2 for partner APIs; per-partner data scoping
Application Services	20 domain microservices (see §6)	Node.js / TypeScript Python/Django (ML & analytics)	Each service owns its DB namespace; no direct cross-service DB access
Event Streaming	Async inter-service communication; payment event processing	Apache Kafka	At-least-once delivery; consumer groups per service; DLQ for failed events
Blockchain	Immutable transaction ledger for all financial events	Hyperledger Fabric 2.5 Golang chaincode	Private permissioned; 3-node Raft; 2-of-3 endorsement; per-group channels
Data — Transactional	All structured operational data	PostgreSQL 15 (Read replica for analytics feeds)	Append-only financial tables; AES-256 at rest; field-level encryption for PII
Data — Analytics	KPI computation, MEL indicators, partner dashboards	BigQuery TensorFlow Lite (credit scoring)	Separate from OLTP; batch-updated via Kafka consumer; no PII in warehouse
Offline Sync	Local data store on Mobile App; conflict-free merge on reconnect	SQLite (on-device) CRDT merge algorithm IWL Sync Engine	Full 8-step meeting works offline; OTP pre-cached; M-Pesa queue on reconnect
Infrastructure	Cloud hosting, CI/CD, secrets, monitoring, alerting	AWS Kenya Region GitHub Actions AWS KMS Grafana / Prometheus / Sentry	Kenya-resident data; 99.5% uptime SLA; automated rollback on failed deploy

4.3 Two-Product Architecture

Product	Primary Users	Channel	Core Purpose
Intellicash Mobile App	VSLA Members, Group Officers, Key-Holders, Village Agents / CBTs	Android & iOS Native App (React Native)	End-to-end VSLA lifecycle management: meetings, passbooks, loans, voting, CBT field work, group formation
Intellicash Web Platform	Partner Officers (NGO, MFI, Govt), Banks & Lenders, IWL Programme Administrators, IWL Loan Officers	Responsive web browser (Next.js)	Portfolio intelligence, MEL reporting, partner administration, IWL loan management, system configuration, API developer portal

SECTION 5

Mobile App — Module Specifications

Screen-by-screen functional requirements for the Intellicash Mobile App

5.1 Navigation Structure

The Mobile App uses a five-tab bottom navigation bar. Tabs and visible actions adapt dynamically to the logged-in user's account type and role. No tab or action belonging to another role may ever be visible, even partially, to a user who does not hold that role.

Tab	Visible To	Key Screens & Actions
☐ Dashboard	All roles	Role-specific overview; Quick Actions (context-sensitive); Recent Transactions list; Group KPI summary; Sync status indicator; Next meeting countdown
☐ Meetings	Officers, Members	Meeting list with status; Start New Meeting (officers); 3-tab session view (Attendance Shares Actions); 8-step enforced workflow; Minutes recorder; OTP activation panel
☐ Members	Officers, Members	Member roster with search/filter; Individual passbook view; Member profiles; KYC status badge; Role assignment (officers only); Credit score view (officers only)
☐ Loans / ☐ CBT	Officers/Members: Loan book Agents: CBT Module	Officers/Members: Loan applications, repayment schedule, disbursement status, guarantor view Agents: Caseload dashboard, visit log, training modules, phase advancement, VAN
... More	All roles (context-specific)	Group Finance (officers); Share-Out Wizard (Treasurer); Reports (officers); Transactions; Settings & Security; Sync status; Account Management; Help & Audio Guides

5.2 IWL 8-Step Meeting Workflow

The Workflow Engine enforces strict procedural order. The Chairperson CANNOT skip, reorder, or reverse any step. The system must make it technically impossible to access Step N+1 before Step N is formally closed. This is not a UX preference — it is a core integrity requirement.



Step	Name	Key Actions Required	Payment Rail	Blockchain Event
1	Opening & Three-Key Security	Three-key OTP unlock + GPS verification for all 3 key-holders. Attendance recorded for all members (present/absent/late). Fines collected from absent and late members.	M-Pesa STK Push (fines)	meeting.opened; otp.verified ×3
2	Minutes Review	Secretary presents previous meeting minutes on screen. Members vote to approve via quick ballot (simple majority). Status set to Approved.	None	minutes.approved
3	Social Fund Round	Secretary calls each member by name. Social Fund contribution collected from each. Grant requests presented and voted on. Approved grants disbursed immediately.	STK Push (contributions); B2C (grants post-vote)	socialfund.collected; grant.disbursed
4	Loan Repayments	Secretary calls each borrower by name. Repayment collected. Borrower OTP-confirms. Passbook entry updated live. Any partial payments flagged as arrears.	M-Pesa STK Push	loan.repayment.received
5	Share Purchase	Secretary calls each member. Member selects 1–5 shares. Digital stamp applied on M-Pesa payment confirmation. Share count visible to all member devices in real time.	M-Pesa STK Push	share.purchased
6	Loan Applications	Member states purpose. Eligibility check (3× savings). Guarantor nomination and acceptance. GA anonymous vote. Disbursement on approval with borrower OTP.	M-Pesa B2C (on GA approval)	loan.disbursed; guarantee.accepted
7	Resolutions & General Votes	Chairperson tables other resolutions (officer elections, constitution changes, external loan approvals, member expulsions). Anonymous ballot. Blockchain-hashed result.	None (governance only)	resolution.passed / .failed
8	Closing	Treasurer announces all fund balances. Three-key seal: key-holders re-confirm session close. Session token expires. Audit log sealed. Webhooks fire to partner Web Platform.	None	meeting.closed; audit.sealed

5.3 CBT / Village Agent Module

Feature	Description	Data Captured	Blockchain?
Caseload Dashboard	All assigned groups with phase, last visit date, quality score, and next required visit date. Colour-coded by risk level.	group_id, phase, last_visit_date, quality_score	No
Group Formation Wizard	Step-by-step guided group formation: member registration, IPRS KYC, officer assignment, key-holder designation, constitution configuration, GPS setup.	All group_accounts and group_members fields	Yes — group.created
Structured Visit Log	GPS-stamped visit record using IWL's 12-indicator quality checklist. Photo upload. VA observations. Group response notes.	visit_id, group_id, agent_id, GPS, timestamp, 12 indicator scores, photos	No — but GPS-stamped
Training Module Delivery	9 IWL training modules delivered via app. Group members follow on own Member Account devices. Completion tracked per member per module.	training_completion per member_id	No
Quality Score Auto-Computation	Score auto-computed from: attendance (20%), passbook accuracy (20%), repayment rate (25%), constitution compliance (20%), visit compliance (15%).	quality_score updated post each visit	No
Phase Advancement	VA signs off on phase advancement in CBT Module. IWL Programme Admin counter-signs on Web Platform. Partner notified via webhook. Group constitution may update with new requirements.	phase_advancement_id, agent_sign_off, admin_sign_off	No
VA Network (VAN)	Peer VA network: register, schedule VAN meetings, share cross-group learning notes, access VA credit fund information.	van_meeting records, learning_notes	No

5.4 Share-Out Wizard

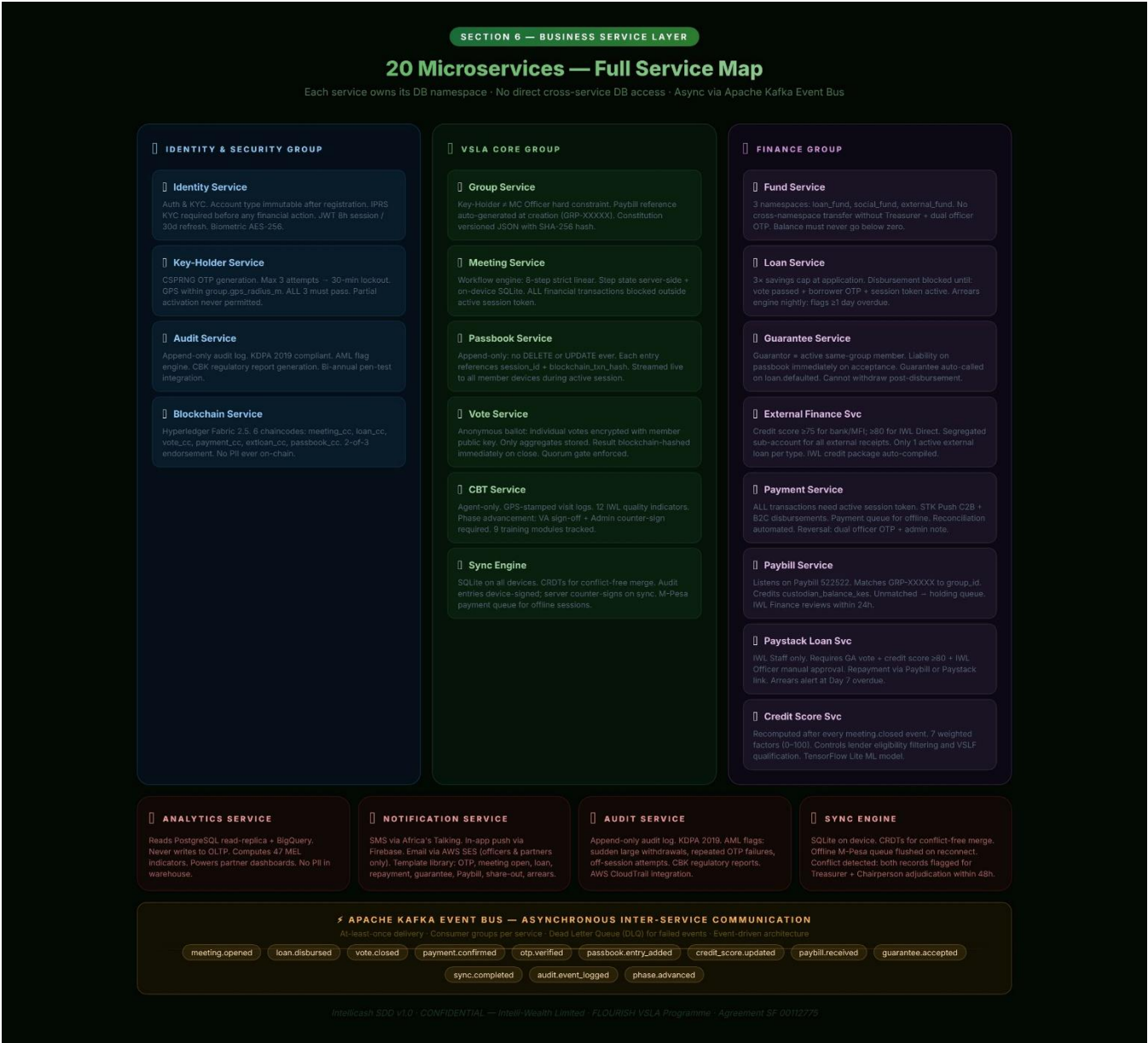
Wizard Step	Action	System Check	Output
1. Pre-Audit Checks	System auto-checks all outstanding loans, liabilities, and guarantees	All loans resolved? All external repaid? All guarantees discharged?	Pass/Fail report shown to Treasurer
2. Interest Calculation	System computes total interest earned on loan fund across full cycle	Interest income = sum of all service charges collected	Interest total displayed to group
3. Share-Out Computation	System computes each member's entitlement: $\text{shares} \times \text{value} + \text{pro-rata interest} - \text{liabilities}$	Validated against total fund balance (must balance to zero)	Per-member entitlement schedule
4. GA Vote on Schedule	Chairperson tables schedule as Resolution; GA votes	Simple majority required	Vote hash recorded on blockchain
5. M-Pesa B2C Batch	Treasurer initiates batch B2C payout to all members	Each transaction queued and dispatched sequentially	M-Pesa refs for each payment
6. Receipt Confirmation	Each member OTP-confirms receipt on Member Account	All members confirmed? Any pending?	Passbook: CYCLE CLOSED
7. Cycle Close	System closes cycle, increments counter, archives passbooks, opens new cycle	All confirmations received	New cycle active; old cycle archived

SECTION 6

Business Service Layer

All 20 microservices: responsibilities, rules, integrations, and events

6.1 Service Architecture Diagram



6.2 Service Specifications

Service	Domain	Key Business Rules (Non-Negotiable)	Integration s	Kafka Events Published
Identity Service	Auth & KYC	Account type is immutable after first registration (member/agent/iwl_staff/partner) . IPRS KYC required before any financial action. Biometric hash stored AES-256 encrypted. JWT expiry: 8h session, 30 days refresh.	IPRS Kenya API Africa's Talking OTP Firebase Auth	user.registered user.kyc_verified user.login user.suspended
Group Service	VSLA Management	Key-Holders CANNOT be MC officers — hard system constraint. Paybill reference auto-generated at group creation (GRP-XXXXX). GPS radius default 50m; configurable by IWL Admin only. Constitution encoded as versioned JSON with SHA-256 hash.	IPRS Kenya GPS/Maps API	group.created group.phase_advanced group.constitution_amended group.activated
Meeting Service	Workflow Engine	8-step linear workflow: cannot skip, reverse, or access Step N+1 before Step N closes. ALL financial transactions blocked outside an active session token. Step state stored server-side and on-device SQLite for offline use.	Key-Holder Service (OTP gate) Passbook Service Blockchain Service	meeting.opened meeting.step_completed meeting.closed meeting.abandoned
Key-Holder Service	Security	OTP generated as CSPRNG 6-digit code. Max 3 attempts per key-holder then 30-min lockout. GPS must be within group.gps_radius_m of meeting location. ALL 3 must pass before session token is issued. No partial activation permitted.	Africa's Talking SMS USSD (fallback OTP) GPS API	otp.sent otp.verified otp.failed otp.expired session.activated
Passbook Service	Ledger	Append-only: no DELETE or UPDATE permitted on any passbook_entries row. Ever. Each entry has: session_id reference (must match active session), blockchain_txn_hash. Passbook streamed live to all member devices during active session.	Blockchain Service Meeting Service	passbook.entry_added passbook.cycle_closed

Service	Domain	Key Business Rules (Non-Negotiable)	Integration s	Kafka Events Published
Fund Service	Treasury	Three fund namespaces: loan_fund, social_fund, external_fund. No cross-namespace transfer without explicit Treasurer action + dual officer OTP. Custodian balance sub-ledger for Paybill-only groups tracked separately. Balance must never go below zero.	Paybill Service Payment Service Blockchain Service	fund.credited fund.debited fund.balance_updated fund.reconciled
Loan Service	Internal Credit	3× savings cap enforced at application (not disbursement). Disbursement blocked until: vote_resolution.result='passed' AND borrower OTP confirmed AND session token active. Arrears engine runs nightly: flags loans overdue by ≥1 day.	Vote Service Payment Service Guarantee Service Blockchain Service	loan.applied loan.approved loan.disbursed loan.repaid loan.arrears_flagged loan.defaulted loan.written_off
Guarantee Service	Credit Guarantee	Guarantor must be active member of same group. Guarantee liability recorded on guarantor passbook immediately on acceptance. On loan.defaulted event: guarantee is automatically called. Guarantor notified via SMS and in-app. Guarantee cannot be withdrawn after disbursement.	Loan Service Passbook Service Notification Service Blockchain Service	guarantee.requested guarantee.accepted guarantee.declined guarantee.active guarantee.called guarantee.discharged
External Finance Service	External Credit & Grants	Credit score ≥75 required for bank/MFI applications; ≥80 for IWL Direct Loans. Segregated sub-account enforced for all external loan receipts. Only 1 active external loan per type at a time. IWL credit data package auto-compiled for lender API. Grant fund is completely ring-fenced.	Credit Score Service Payment Service Paystack Loan Service KCB Buni API Blockchain Service	extloan.applied extloan.approved extloan.disbursed extloan.repayment_received extloan.defaulted extloan.repaid grant.received grant.disbursed
Payment Service	Payments	ALL transactions must have active session token. M-Pesa STK Push for C2B (member-to-group). B2C for disbursements (group-to-member). Payment queue for offline sessions — executed in sequence on reconnect. Reconciliation automated. Reversal requires dual officer OTP + admin audit note.	M-Pesa Daraja v3 Paystack KCB Buni API Kafka (async events)	payment.initiated payment.confirmed payment.failed payment.queued payment.reversed payment.reconciled

Service	Domain	Key Business Rules (Non-Negotiable)	Integration s	Kafka Events Published
Paybill Service	IWL Custodian Savings	Listens on M-Pesa webhook for Business No. 522522. Matches account reference (GRP-XXXXX) to group_id. Credits custodian_balance_kes. Unmatched refs → holding queue. IWL Finance reviews within 24h. Reversal handling via Daraja webhook.	M-Pesa Daraja v3 (Paybill C2B webhook) Fund Service Notification Service Blockchain Service	paybill.received paybill.matched paybill.unmatched paybill.allocated paybill.reversed
Paystack Loan Service	IWL Direct Lending	Only IWL Loan Officers (iwl_staff account type) can initiate disbursements. Group must have passed GA vote. Paystack sub-account per group for fund segregation. Repayment via Paybill or Paystack link. Arrears alert at Day 7 overdue.	Paystack API External Finance Service Notification Service Blockchain Service	iwl_loan.applied iwl_loan.approved iwl_loan.rejected iwl_loan.disbursed iwl_loan.repayment_received iwl_loan.arrears_flagged
Vote Service	Governance	Anonymous ballot: individual votes encrypted with member public key. Only aggregate stored in plain text. No officer, admin, or developer can view individual votes. Ever. Result blockchain-hashed immediately on close. Quorum gate: vote cannot close if minimum voters not reached.	Blockchain Service Notification Service	vote.opened vote.cast vote.closed resolution.passed resolution.failed resolution.quorum_invalid
Credit Score Service	Credit Intelligence	Score recomputed after every meeting.closed event. Factors: repayment 25%, savings 20%, attendance 15%, compliance 15%, social fund 10%, cycle age 10%, security 5%. Score used for lender eligibility filtering and VSLF qualification.	Meeting Service Loan Service Fund Service Analytics Service	credit_score.updated credit_score.threshold_crossed
CBT Service	Training & Field Ops	Agent account type only. Visit log must be GPS-stamped. Phase advancement requires VA sign-off + IWL Admin counter-sign. Quality score uses 12 IWL indicators. Meeting cannot proceed in Intensive Phase Sessions 1–9 without CBT check-in gate.	Group Service GPS API Photo upload (AWS S3) Notification Service	visit.logged training.module_completed phase.advancement_request phase.advanced van.meeting_scheduled

Service	Domain	Key Business Rules (Non-Negotiable)	Integrations	Kafka Events Published
Blockchain Service	Immutable Ledger	Hyperledger Fabric 2.5. 6 chaincodes: meeting_cc, loan_cc, vote_cc, payment_cc, extloan_cc, passbook_cc. Async write via Kafka consumer. Confirmation webhook updates blockchain_txn_hash on originating record. 2-of-3 endorsement policy. No PII ever written to chain.	Hyperledger Fabric network Kafka (consumer) All financial services (webhook callbacks)	blockchain.committed blockchain.commit_failed blockchain.conflict_detected
Analytics Service	Business Intelligence	Reads from PostgreSQL read-replica + BigQuery. Never writes to OLTP DB. Computes 47 MEL indicators. Powers partner Web Platform dashboards. Runs credit scoring model (TensorFlow Lite). Exports on schedule or on-demand via API.	BigQuery Kafka (consumer) TensorFlow Lite	analytics.kpi_computed analytics.mel_report_generated
Audit Service	Compliance	Append-only audit log. KDP A 2019 compliant. AML flag engine: flags unusual transaction patterns (large sudden withdrawals, repeat failed OTPs, off-session attempts). Regulatory report generation (CBK format). Bi-annual penetration test integration.	All services (Kafka consumer) AWS CloudTrail	audit.event_logged audit.aml_flagged audit.security_alert audit.report_generated
Notification Service	Communications	SMS via Africa's Talking. In-app push via Firebase. Email via AWS SES (officers and partners only). Template library: OTP, meeting open, loan approved, repayment reminder, guarantee called, Paybill received, share-out scheduled, arrears alert.	Africa's Talking Firebase Cloud Messaging AWS SES	notification.sent notification.failed notification.delivered
Sync Engine	Offline-First	SQLite on all mobile devices. CRDTs for conflict-free merge. Audit entries locally signed with device key; counter-signed by server on sync. Payment queue for offline M-Pesa events. If conflict detected post-sync: both records flagged for Treasurer + Chairperson adjudication within 48h.	All mobile-facing services IWL Sync Protocol	sync.completed sync.conflict_detected sync.payment_queue_flushed

SECTION 7

Database Schema

Core data entities, relationships, and constraints

The Intellicash database uses PostgreSQL 15 as the primary transactional store. All financial tables are append-only — no DELETE or UPDATE is permitted on any row that represents a completed financial event. Schema is partitioned by domain namespace. All PII fields are encrypted at rest using AES-256. The blockchain_txn_hash column on financial tables is a foreign reference to the Hyperledger Fabric ledger entry.

Primary Key (PK) = UUID, system-generated, immutable. Foreign Key (FK) = cross-table reference. All timestamps are TIMESTAMPTZ (UTC). All monetary values are DECIMAL(14,2) in KES unless stated.

7.1 Identity & Account Domain

users

Column	Type / Constraint	Key	Description
user_id	UUID NOT NULL	PK	System-generated GUID. Immutable after creation.
phone_number	VARCHAR(15) UNIQUE		Primary identifier; Kenya format +254XXXXXXXXX. AES-256 encrypted.
national_id	VARCHAR(20)		IPRS-verified National ID. AES-256 encrypted at rest.
full_name	VARCHAR(120)		As per IPRS record.
date_of_birth	DATE		
gender	CHAR(1)		M / F / O
biometric_hash	VARCHAR(512)		AES-256 encrypted fingerprint/face hash for biometric login.
account_type	ENUM('member','agent','iwl_staff','partner')		Set at first registration. Immutable. Requires Admin change process.
kyc_status	ENUM('pending','verified','failed')		Updated by IPRS webhook. financial actions blocked unless 'verified'.
preferred_language	ENUM('sw','en') DEFAULT 'sw'		Kiswahili default.

Column	Type / Constraint	Key	Description
created_at	TIMESTAMPZ NOT NULL DEFAULT NOW()		Immutable.
last_login_at	TIMESTAMPZ		
is_active	BOOLEAN DEFAULT TRUE		FALSE = suspended. Cannot log in or conduct any transaction.
device_key	VARCHAR(512)		Encrypted public key for offline audit entry signing.

■ group_accounts

Column	Type / Constraint	Key	Description
group_id	UUID NOT NULL	PK	Immutable.
group_name	VARCHAR(120) NOT NULL		
paybill_reference	VARCHAR(20) UNIQUE NOT NULL		Auto-generated: GRP-XXXXX. Used as M-Pesa Paybill account reference.
has_bank_account	BOOLEAN DEFAULT FALSE		Determines fund routing: Paybill custodian vs. direct bank EFT.
bank_account_no	VARCHAR(30)		AES-256 encrypted. NULL if has_bank_account=FALSE.
bank_name	VARCHAR(80)		
custodian_balance_kes	DECIMAL(14,2) DEFAULT 0		IWL custodian sub-account balance. Only relevant when has_bank_account=FALSE.
gps_lat	DECIMAL(9,6) NOT NULL		Registered meeting location latitude.
gps_lng	DECIMAL(9,6) NOT NULL		Registered meeting location longitude.

Column	Type / Constraint	Key	Description
gps_radius_m	INTEGER DEFAULT 50		Allowed GPS radius for meeting activation. Configurable by IWL Admin only.
lifecycle_phase	ENUM('mobilisation','intensive','development','maturity','graduated') NOT NULL		
current_cycle	INTEGER DEFAULT 1		Increments at each Action Audit / Share-Out completion.
credit_score	DECIMAL(5,2) DEFAULT 0		IWL Credit Score 0–100. Recomputed post each meeting.closed event.
constitution_version	INTEGER DEFAULT 1		Increments on each amendment.
constitution_hash	VARCHAR(64) NOT NULL		SHA-256 hash of current constitution JSON. Stored immutably at each version.
assigned_agent_id	UUID	FK	→ users.user_id (agent account type).
status	ENUM('pending','active','suspended','graduated','closed') DEFAULT 'pending'		'active' set by IWL Admin after agent submits and Admin counter-signs.
created_at	TIMESTAMPTZ NOT NULL DEFAULT NOW()		
blockchain_creation_hash	VARCHAR(128)		Hyperledger Fabric TX hash for group.created event.

■ group_members

Column	Type / Constraint	Key	Description
membership_id	UUID NOT NULL	PK	
group_id	UUID NOT NULL	FK	→ group_accounts.group_id

Column	Type / Constraint	Key	Description
user_id	UUID NOT NULL	FK	→ users.user_id. UNIQUE constraint with group_id.
role	ENUM('member','chairperson','secretary','treasurer','money_counter_1','money_counter_2','keyholder_1','keyholder_2','keyholder_3') NOT NULL		CONSTRAINT: keyholder roles cannot overlap with chairperson/secretary/treasurer.
joined_date	DATE NOT NULL		
status	ENUM('active','inactive','expelled','withdrawn') DEFAULT 'active'		
share_balance	INTEGER DEFAULT 0		Total cumulative share stamps purchased.
share_value kes	DECIMAL(14,2) DEFAULT 0		Computed: share_balance × group share denomination. Updated after each share purchase.

7.2 Meeting & OTP Domain

■ meeting_sessions

Column	Type / Constraint	Key	Description
session_id	UUID NOT NULL	PK	
group_id	UUID NOT NULL	FK	→ group_accounts.group_id
session_token_hash	VARCHAR(128) NOT NULL		Hashed JWT. Never store raw token.
started_at	TIMESTAMPTZ		Set when all 3 OTPs verified and GPS compliant.
closed_at	TIMESTAMPTZ		NULL while active. Set at Step 8 close.
gps_lat_actual	DECIMAL(9,6)		GPS at activation — compliance audit.
gps_lng_actual	DECIMAL(9,6)		

Column	Type / Constraint	Key	Description
step_current	INTEGER DEFAULT 0		0 = OTP pending. 1–8 = active step. 9 = closed. Cannot decrement.
attendance_count	INTEGER DEFAULT 0		Members confirmed present at Step 1.
total_fines_collected	DECIMAL(14,2) DEFAULT 0		Aggregate fines at Step 1.
total_shares_purchased	DECIMAL(14,2) DEFAULT 0		Aggregate share value at Step 5.
total_loans_disbursed	DECIMAL(14,2) DEFAULT 0		Aggregate at Step 6.
blockchain_open_hash	VARCHAR(128)		Hyperledger TX hash for meeting.opened event.
blockchain_close_hash	VARCHAR(128)		Hyperledger TX hash for meeting.closed event.

■ otp_events

Column	Type / Constraint	Key	Description
otp_id	UUID NOT NULL	PK	
session_id	UUID NOT NULL	FK	→ meeting_sessions.session_id
keyholder_user_id	UUID NOT NULL	FK	→ users.user_id
keyholder_position	SMALLINT NOT NULL		1, 2, or 3. Corresponds to keyholder_1/2/3 role.
otp_hash	VARCHAR(64) NOT NULL		SHA-256 of the OTP sent. Raw OTP NEVER stored.
sent_at	TIMESTAMPTZ NOT NULL		
verified_at	TIMESTAMPTZ		NULL if not yet verified.
attempts	SMALLINT DEFAULT 0		Incremented on each failed attempt. Hard stop at 3.
gps_lat	DECIMAL(9,6)		Key-Holder device GPS at verification attempt.
gps_lng	DECIMAL(9,6)		
gps_compliant	BOOLEAN		TRUE if within group.gps_radius_m.

Column	Type / Constraint	Key	Description
status	ENUM('sent','verified','failed','expired','locked') NOT NULL DEFAULT 'sent'		
delivery_channel	ENUM('sms','ussd') DEFAULT 'sms'		USSD used as fallback if SMS fails.

7.3 Loan & Credit Guarantee Domain

■ member_loans

Column	Type / Constraint	Key	Description
loan_id	UUID NOT NULL	P K	
group_id	UUID NOT NULL	F K	→ group_accounts.group_id
borrower_user_id	UUID NOT NULL	F K	→ users.user_id
session_id	UUID NOT NULL	F K	→ meeting_sessions.session_id (meeting of approval)
loan_type	ENUM('internal','external_mfi','external_bank','iwl_direct') NOT NULL		
principal_kes	DECIMAL(14,2) NOT NULL		Amount approved by GA vote.
service_charge_rate_pct	DECIMAL(5,2) NOT NULL		Flat % per month. From group constitution at time of application.
term_months	SMALLINT NOT NULL		
purpose	TEXT NOT NULL		Member-stated purpose, recorded verbatim.
eligibility_multiplier_at_application	DECIMAL(5,2) NOT NULL		Borrower's share_value × 3 at time of application. Immutable snapshot.

Column	Type / Constraint	Key	Description
vote_resolution_id	UUID NOT NULL	FK	→ vote_resolutions.resolution_id. Must be status='passed'.
status	ENUM('pending_vote','approved','disbursement_pending','disbursed','repaid','defaulted','written_off') NOT NULL DEFAULT 'pending_vote'		
disbursed_at	TIMESTAMPTZ		Set on M-Pesa B2C confirmation.
disbursement_mpesa_ref	VARCHAR(30)		M-Pesa B2C transaction reference.
paystack_txn_ref	VARCHAR(60)		For IWL Direct Loans via Paystack.
repaid_at	TIMESTAMPTZ		Set when final repayment received.
blockchain_txn_hash	VARCHAR(128)		Hyperledger Fabric TX hash for loan.disbursed event.

■ loan_guarantees

Column	Type / Constraint	Key	Description
guarantee_id	UUID NOT NULL	PK	
loan_id	UUID NOT NULL	FK	→ member_loans.loan_id
guarantor_user_id	UUID NOT NULL	FK	→ users.user_id. Must be active member of same group as borrower.
group_id	UUID NOT NULL	FK	→ group_accounts.group_id
liability_amount_ke	DECIMAL(14,2) NOT NULL		Pro-rata share of loan if multiple guarantors; full loan if sole guarantor.

Column	Type / Constraint	Key	Description
nominated_at	TIMESTAMPTZ NOT NULL		When borrower nominated this guarantor.
status	ENUM('requested','accepted','declined','active','called','discharged') NOT NULL DEFAULT 'requested'		'called' = loan defaulted and guarantee activated.
accepted_at	TIMESTAMPTZ		When guarantor accepted in-app.
called_at	TIMESTAMPTZ		When guarantee was called (loan.defaulted event).
discharged_at	TIMESTAMPTZ		When loan fully repaid and guarantee released.
blockchain_txn_hash	VARCHAR(128)		Immutable record of guarantee.accepted event on blockchain.

■ loan_repayments

Column	Type / Constraint	Key	Description
repayment_id	UUID NOT NULL	PK	
loan_id	UUID NOT NULL	FK	→ member_loans.loan_id
session_id	UUID	FK	→ meeting_sessions.session_id. NULL only for Paybill repayments outside meeting.
amount_principal_kes	DECIMAL(14,2) NOT NULL		
amount_service_charge_kes	DECIMAL(14,2) NOT NULL		
mpesa_txn_ref	VARCHAR(30)		M-Pesa STK Push confirmation reference.
paybill_txn_id	UUID	FK	→ paybill_transactions.paybill_txn_id. For Paybill repayments.
paid_at	TIMESTAMPTZ NOT NULL		

Column	Type / Constraint	Key	Description
otp_confirmed	BOOLEAN DEFAULT FALSE		Borrower OTP receipt confirmation.
repayment_type	ENUM('scheduled','partial','early','catch_up') DEFAULT 'scheduled'		
blockchain_txn_hash	VARCHAR(128)		Hyperledger Fabric TX hash for loan.repayment.received event.

7.4 External Finance Domain

■ external_loan_applications

Column	Type / Constraint	Key	Description
application_id	UUID NOT NULL	P K	
group_id	UUID NOT NULL	F K	→ group_accounts. group_id
lender_type	ENUM('bank','mfi','iwl_direct') NOT NULL		
lender_partner_id	UUID	F K	→ partners.partner_id. NULL for IWL Direct loans.
amount_requested_kes	DECIMAL(14,2) NOT NULL		
amount_approved_kes	DECIMAL(14,2)		May differ from requested if lender makes counter-offer.
purpose	TEXT NOT NULL		
credit_score_at_application	DECIMAL(5,2) NOT NULL		Immutable snapshot of IWL Credit Score at time of submission.
vote_resolution_id	UUID NOT NULL	F K	→ vote_resolutions. resolution_id

Column	Type / Constraint	Key	Description
status	ENUM('draft','pending_vote','submitted','under_review','approved','disbursed','rejected','defaulted','restructured','repaid') NOT NULL DEFAULT 'draft'		
disbursement_route	ENUM('bank_eft','mpesa_b2c','iwl_custodian','paystack') NOT NULL		
paystack_txn_ref	VARCHAR(60)		For IWL Direct Loans.
iwl_officer_user_id	UUID	FK	→ users.user_id (iwl_staff type). Required for IWL Direct Loans.
submitted_at	TIMESTAMPTZ		
disbursed_at	TIMESTAMPTZ		
blockchain_txn_hash	VARCHAR(128)		Hyperledger TX hash for extloan.disbursed event.

■ paybill_transactions

Column	Type / Constraint	Key	Description
paybill_txn_id	UUID NOT NULL	PK	
group_id	UUID	FK	→ group_accounts.group_id. NULL if account reference unmatched.
mpesa_transaction_id	VARCHAR(30) UNIQUE NOT NULL		M-Pesa webhook TransactionID. Deduplicated.
sender_phone	VARCHAR(15)		AES-256 encrypted. Phone of paying sender.
amount_kes	DECIMAL(14,2) NOT NULL		

Column	Type / Constraint	Key	Description
account_reference	VARCHAR(20) NOT NULL		GRP-XXXXXX as entered by sender. May be malformed for unmatched.
received_at	TIMESTAMPTZ NOT NULL		M-Pesa webhook timestamp.
matched	BOOLEAN DEFAULT FALSE		TRUE if account_reference resolved to a valid group_id.
allocation_fund	ENUM('loan_fund','social_fund','external_loan_repayment','iwl_loan_repayment','unallocated')		Set by Treasurer.
allocated_at	TIMESTAMPTZ		NULL until Treasurer allocates.
allocated_by_user_id	UUID	FK	→ users.user_id (Treasurer)
reversed	BOOLEAN DEFAULT FALSE		TRUE if M-Pesa reversal webhook received.
blockchain_txn_hash	VARCHAR(128)		Hyperledger TX hash for paybill.received event.

7.5 Governance Domain

■ vote_resolutions

Column	Type / Constraint	Key	Description
resolution_id	UUID NOT NULL	PK	
group_id	UUID NOT NULL	FK	→ group_accounts.group_id
session_id	UUID NOT NULL	FK	→ meeting_sessions.session_id

Column	Type / Constraint	Key	Description
resolution_type	ENUM('internal_loan','external_loan','iwl_direct_loan','grant_application','social_fund_grant','officer_election','constitution_amendment','member_expulsion','vslf_membership','fine_waiver','minutes_approval','share_out_schedule','other') NOT NULL		
motion_text	TEXT NOT NULL		Full text of the motion as tabled. Immutable.
quorum_required_pct	DECIMAL(5,2) NOT NULL		75.00 or 80.00 based on resolution type.
total_eligible_voters	INTEGER NOT NULL		Members present at meeting Step 1 attendance.
votes_yes	INTEGER DEFAULT 0		Aggregate only. Individual votes never stored in plain text.
votes_no	INTEGER DEFAULT 0		
votes_abstain	INTEGER DEFAULT 0		
result	ENUM('passed','failed','invalid_quorum','pending') NOT NULL DEFAULT 'pending'		'invalid_quorum' if fewer than 75% of eligible voters voted.
resolved_at	TIMESTAMP TZ		Set when vote closes.
blockchain_result_hash	VARCHAR(128)		SHA-256(resolution_id+yes+no+abstain+result+resolved_at). Written to Hyperledger vote_cc chaincode.

■ passbook_entries

Column	Type / Constraint	Key	Description
entry_id	UUID NOT NULL	P K	
group_id	UUID NOT NULL	F K	→ group_accounts.group_id
member_user_id	UUID NOT NULL	F K	→ users.user_id
session_id	UUID NOT NULL	F K	→ meeting_sessions.session_id. EVERY entry must have a session.
entry_type	ENUM('share_purchase','loan_disbursed','loan_repayment','social_fund_contribution','social_fund_grant','fine','share_out','external_loan_disbursed','external_loan_repayment','guarantee_called','cycle_opened','cycle_closed') NOT NULL		
amount_kes	DECIMAL(14,2) NOT NULL		Positive = credit; Negative = debit.
running_balance_kes	DECIMAL(14,2) NOT NULL		Member's cumulative balance at this entry. Append-only.
reference_id	UUID		Optional: loan_id, guarantee_id, etc. for cross-referencing.
narrative	VARCHAR(255) NOT NULL		Human-readable description.
recorded_at	TIMESTAMPTZ NOT NULL DEFAULT NOW()		Immutable.

Column	Type / Constraint	Key	Description
blockchain_txn_hash	VARCHAR(128)		Hyperledger TX hash for passbook. entry_added event.

SECTION 8

Blockchain Architecture

Hyperledger Fabric — immutable transaction ledger

All financial transactions, meeting activations, vote results, loan events, and guarantee agreements are written to a private, permissioned Hyperledger Fabric blockchain. This provides tamper-proof, cryptographically immutable records independently auditable by IWL, partners, regulators, and courts. No blockchain record can ever be altered — not by IWL administrators, developers, or database operations.

SECTION 8 — BLOCKCHAIN ARCHITECTURE

Hyperledger Fabric — Immutable Ledger

Private permissioned blockchain · All financial events cryptographically recorded · No record can ever be altered — not by IWL admins, developers, or database operations

BLOCKCHAIN TYPE Hyperledger Fabric 2.5 Private Permissioned	CONSENSUS Raft Ordering Service Crash Fault Tolerant	ENDORSEMENT POLICY 2-of-3 peer nodes must endorse each tx	PEER NODES 3x IWL nodes AWS Cape Town + Kenya
PER-GROUP CHANNELS Private data collection Groups cannot see each other	THROUGHPUT TARGET 1,000 TPS ~2s block finality	OFF-CHAIN POLICY PostgreSQL stores full payload Chain stores hash + key fields	AUDIT INTERFACE Hyperledger Explorer Read-only for CBK, partners

6 CHAINCODE DOMAINS

meeting_cc Meeting Lifecycle meeting.opened meeting.closed KEY FIELDS ON-CHAIN (NO PII) group_id_hash · session_id · otp_hashes[3] · gps_hash · step_completed · timestamp	loan_cc Member Loans & Guarantees loan.disbursed loan.repaid guarantee.accepted guarantee.called KEY FIELDS ON-CHAIN (NO PII) loan_id · borrower_hash · amount · guarantor_hashes[] · vote_hash · mpesa_ref · timestamp	vote_cc General Assembly Votes vote.opened resolution.passed resolution.failed KEY FIELDS ON-CHAIN (NO PII) resolution_id · group_id_hash · yes_count · no_count · abstain_count · result · quorum_pct · timestamp
payment_cc Payments & Paybill payment.confirmed payment.reversed paybill.received paybill.allocated KEY FIELDS ON-CHAIN (NO PII) payment_id · group_id_hash · amount · mpesa_txn_id · allocation_fund · timestamp	extloan_cc External & IWL Loans extloan.disbursed iwl_loan.disbursed extloan.repaid KEY FIELDS ON-CHAIN (NO PII) application_id · lender_id_hash · group_id_hash · amount · disbursement_route · paystack_ref · vote_hash · timestamp	passbook_cc Passbook Entries passbook.entry_added passbook.cycle_closed KEY FIELDS ON-CHAIN (NO PII) entry_id · member_hash · group_id_hash · entry_type · amount · running_balance · session_id · timestamp

ABSOLUTE RULE: Member PII (name, ID, phone number) is NEVER written to the blockchain — KCPA 2019 compliance. Only hashes on-chain.

ON-CHAIN VS OFF-CHAIN DATA POLICY

On-Chain (Hyperledger Fabric Ledger)	Off-Chain (PostgreSQL — AES-256 Encrypted)
Transaction event hashes Full event hash + type + key fields + timestamp	Full transaction records All fields, narratives, references — searchable and reportable
Loan amounts Amount only — no beneficiary name. Auditability without PII exposure.	Member PII phone_number, national_id, full_name, biometric_hash — AES-256 encrypted at rest
Vote results Aggregate counts + resolution type + result only. Individual votes: technically impossible to recover.	Full resolution records Motion text, quorum data, meeting context — full accountability trail
OTP verification hashes SHA-256 hash of each OTP — not the OTP itself. Proves meeting unlocked correctly.	Full OTP events table Attempt counts, delivery channel, GPS coordinates — compliance audit
GPS coordinate hashes SHA-256 hash of lat/long pair. Location proof without tracking individuals.	Exact GPS coordinates Stored in meeting_sessions for compliance audit — never on-chain
Guarantee agreements guarantee_id hash + liability amount. Immutable proof of commitment.	Full passbook entries Narrative, reference_id cross-links, running balance — member-readable

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8.1 Hyperledger Fabric Network Configuration

Parameter	Configuration	Rationale
Blockchain Type	Hyperledger Fabric 2.5 (private permissioned network)	Regulated financial data; no public exposure; KDPA 2019 compatible; enterprise-grade
Consensus Mechanism	Raft ordering service (crash fault tolerant)	Suitable for AWS Kenya Region; tolerates up to $n/2$ node failures
Endorsement Policy	2-of-3 peer nodes must endorse each transaction	Prevents single-node manipulation; remains available with 1 node offline
Peer Nodes	3 × IWL peer nodes (AWS Africa — Cape Town + Kenya Region)	Geographic redundancy; Kenyan data sovereignty for at least 2 nodes
Channels	Separate private data collection per group	Groups cannot see each other's transaction data
Chaincode Language	Go (Golang)	Hyperledger recommended; performant; type-safe
Off-Chain Storage	PostgreSQL stores full payload; blockchain stores event type + hash + key fields only	Minimises on-chain data storage; PII stays off-chain always
Transaction Throughput	Target 1,000 TPS	Sufficient for all Kenya + East Africa VSLA transaction volume at scale
Audit Interface	Hyperledger Explorer (read-only web interface)	IWL Admin, CBK regulator, authorised partner can independently verify any transaction
Block Finality	~2 seconds (Raft consensus)	Acceptable for VSLA meeting context; passbook entry visible within 5s

8.2 Chaincode Domains

Chaincode	Domain	Events Written	Key Fields On-Chain (no PII)	Trigger
meeting_cc	Meeting Lifecycle	meeting.opened meeting.closed	group_id_hash, session_id, otp_hashes[3], gps_hash, step_completed, timestamp	All 3 OTPs verified; meeting closed at Step 8

Chaincode	Domain	Events Written	Key Fields On-Chain (no PII)	Trigger
loan_cc	Member Loans & Guarantees	loan.disbursed loan.repaid loan.defaulted guarantee.accepted guarantee.called guarantee.discharged	loan_id, borrower_hash, amount, guarantor_hashes[], vote_hash, mpesa_ref, timestamp	Loan disbursement confirmed; repayment received; guarantee events
vote_cc	General Assembly Votes	vote.opened resolution.passed resolution.failed	resolution_id, group_id_hash, yes_count, no_count, abstain_count, result, quorum_pct, timestamp	Vote closed; result computed
payment_cc	Payments & Paybill	payment.confirmed payment.reversed paybill.received paybill.allocated	payment_id, group_id_hash, amount, mpesa_txn_id, paybill_txn_id, allocation_fund, timestamp	Payment webhook confirmed; Paybill received and matched
extloan_cc	External Loans & IWL Direct	extloan.disbursed iwl_loan.disbursed extloan.repaid	application_id, lender_id_hash, group_id_hash, amount, disbursement_route, paystack_ref, vote_hash, timestamp	External loan or IWL loan disbursed
passbook_cc	Passbook Entries	passbook.entry_added passbook.cycle_closed	entry_id, member_hash, group_id_hash, entry_type, amount, running_balance, session_id, timestamp	Each passbook entry written during active session; cycle close

8.3 On-Chain vs. Off-Chain Data Policy

Data Category	On-Chain (Fabric Ledger)	Off-Chain (PostgreSQL)	Compliance Rationale
Transaction event hashes	✓ Full event hash + type + key fields + timestamp	✓ Full record with all fields	Immutability on-chain; searchable/reportable off-chain
Member PII (name, ID, phone)	✗ NEVER on-chain	✓ AES-256 encrypted in PostgreSQL	KDPA 2019 right to erasure via anonymisation

Data Category	On-Chain (Fabric Ledger)	Off-Chain (PostgreSQL)	Compliance Rationale
Loan amounts	✓ Amount only, no beneficiary name	✓ Full loan record with borrower reference	Auditability without personal data exposure on ledger
Vote results	✓ Aggregate counts + resolution type + result	✓ Full resolution record	Individual vote privacy; only aggregates public
OTP verification hashes	✓ Hash of each OTP (not the OTP itself)	✓ otp_events table	Proves meeting unlocked correctly without exposing OTP
Guarantee agreements	✓ guarantee_id hash + liability amount	✓ Full loan_guarantees record	Immutable proof of commitment without PII
GPS coordinates	✓ SHA-256 hash of lat/long pair	✓ Exact coordinates in meeting_sessions	Location proof without tracking individual positions
Paybill receipts	✓ paybill_txn_id + amount + group_hash + timestamp	✓ Full paybill_transactions record	Immutable custody proof for IWL Custodian account
Passbook entries	✓ entry_id + entry_type + amount + member_hash + session_id	✓ Full passbook_entries record with narrative	Tamper-proof ledger; PII stays off-chain

SECTION 9

Security Framework

Five-layer anti-manipulation model, separation of duties, and data protection

9.1 IWL Five-Layer Anti-Manipulation Framework

Layer	Physical VSLA Equivalent	Digital Implementation	Threat Prevented
1 — Identity	Members know each other personally	IPRS KYC + biometric login + group member photo verification at registration	Impersonation; phantom member registration; fake identity
2 — Three-Key Meeting Lock	Three padlocks on cashbox; three separate physical keys	3-key OTP (6-digit SMS/USSD) + GPS geofence (50m); all three must pass; meeting hard-blocked if any fail	Meeting opened without all key-holders; single-officer control of funds; remote fraud
3 — Per-Transaction OTP	Member physically counts and receives money in front of group	Every M-Pesa STK Push debit and B2C credit requires a member OTP confirmation before execution	Silent unauthorised deductions; Treasurer unilateral fraud; ghost disbursements
4 — Real-Time Group Visibility	All transactions done in front of all members simultaneously	Live meeting ledger streamed to all member devices during active session in real time	Secret transactions; retroactive record manipulation; officer collusion
5 — Immutable Ledger	Passbook entries are permanent (red pen only for cancellations)	Append-only, blockchain-backed event ledger; no DELETE/UPDATE at any system layer, ever	Retroactive data fraud; cover-up of embezzlement; disputed transaction manipulation

9.2 System-Enforced Separation of Duties

Rule	Enforcement Method	Consequence of Violation Attempt
Chairperson cannot approve their own loan application	Loan Service: auto-assigns a different officer as approver if applicant is Chairperson	Hard block — system rejects action with error message; audit event logged
Chairperson cannot cast a deciding vote on their own motion	Vote Service: Chairperson's vote is accepted but recusal option is presented; result validated without their vote if outcome-determinative	Audit flag raised; IWL Admin notified if pattern repeated
Treasurer cannot confirm their own loan repayment	Loan Service: repayment confirmation requires a different officer OTP	Hard block; audit event logged

Rule	Enforcement Method	Consequence of Violation Attempt
Treasurer cannot disburse to themselves without dual OTP	Payment Service: any B2C where recipient matches Treasurer's phone triggers dual officer OTP requirement	Hard block; additional audit event; IWL Admin alert
Key-Holders cannot be MC officers	Group Service: enforced at role assignment; checked at every OTP event	Hard block at formation and at any attempted role change
Loan cancellation requires dual officer sign-off	Loan Service: Chairperson + Treasurer both must provide OTP to cancel	Hard block; single-officer cancellation attempt logged as security event
IWL system admins cannot create or modify group financial records	API Gateway: admin JWT scopes exclude all write paths to group financial data	403 Forbidden at gateway; audit event logged; IWL security team alerted
Partner API access is read-only	API Gateway: partner OAuth2 tokens have no write scopes; all write endpoints return 403 to partner tokens	403 Forbidden; logged; webhook notification to partner admin on repeated attempts
Vote anonymity: no user can view individual ballots	Vote Service: individual votes encrypted client-side with member public key before transmission; server never receives plaintext ballot	Decryption technically impossible without member's private key

9.3 Data Protection & KDPA 2019 Compliance

Requirement	Implementation	Standard / Regulation
Encryption at rest	AES-256 for all PII fields (phone, national_id, biometric_hash, bank_account_no, sender_phone). Database-level encryption for all tables.	KDPA 2019 §25; IWL Data Protection Policy
Encryption in transit	TLS 1.3 for all API connections; certificate pinning on Mobile App; HSTS on Web Platform.	KDPA 2019 §25; CBK Cybersecurity Guidelines
Right to erasure	Anonymisation (not deletion) for transactional integrity. PII replaced with 'ANONYMISED' string. Financial records retained as required. Blockchain hashes remain (no PII on chain).	KDPA 2019 §26; CBK 7-year financial record retention
Data minimisation	Only collect fields necessary for VSLA operation. No marketing data. No behavioural analytics on members.	KDPA 2019 §25(c)

Requirement	Implementation	Standard / Regulation
Consent at registration	Explicit opt-in consent form at account creation. Language: Kiswahili and English. Consent record stored immutably.	KDPA 2019 §30
DPO registration	IWL-registered Data Protection Officer with ODPC Kenya. Contact displayed in app.	KDPA 2019 §24
Breach notification	Incident Response Plan: system isolation within 1h; ODPC notification within 72h; affected member SMS within 24h of notification.	KDPA 2019 §43; ODPC Guidelines
Data residency	All operational data stored in AWS regions with at least one Kenya-resident node.	CBK Data Residency Guidelines 2022
Access controls	Role-based access control (RBAC) at API Gateway level. Principle of least privilege. IWL staff access audited monthly.	KDPA 2019 §25; ISO 27001
Penetration testing	Bi-annual third-party penetration testing. Results reported to IWL Board. Critical findings resolved within 30 days.	CBK Cybersecurity Guidelines; IWL Security Policy

SECTION 10

Payments & External Integrations

Payment rails, IWL Paybill, Paystack, and all third-party integrations

Intellicash is a payment pass-through. The platform never holds member funds as a financial institution. All funds flow through licensed payment providers (M-Pesa, Paystack, KCB Buni) or IWL's Custodian Account under IWL's operating licence. No financial transaction is permitted outside a Three-Key unlocked meeting session, except: Paybill deposits, Paystack repayments, and external loan repayments (which occur on scheduled dates, not meeting dates).

10.1 Payment Transaction Map

Transaction Type	Direction	Rail	Meeting Step / Trigger	Member OTP Required?	Blockchain Event
Fine collection	Member → Loan Fund	M-Pesa STK Push	Step 1 — Opening	Yes	payment.confirmed
Social Fund contribution	Member → Social Fund	M-Pesa STK Push	Step 3 — Social Fund	Yes	payment.confirmed
Social Fund grant disbursement	Social Fund → Member	M-Pesa B2C	Step 3 — post-GA vote	Yes (recipient)	payment.confirmed; grant.disbursed
Loan repayment	Member → Loan Fund	M-Pesa STK Push	Step 4 — Loan Repayments	Yes	loan.repayment.received
Share purchase	Member → Loan Fund	M-Pesa STK Push	Step 5 — Share Purchase	Yes	share.purchased
Internal loan disbursement	Loan Fund → Member	M-Pesa B2C	Step 6 — post-GA vote	Yes (borrower)	loan.disbursed
External loan repayment (per member)	Member → External Fund	M-Pesa STK Push	Post-meeting scheduled date	Yes	extloan.repayment_received
External loan receipt (bank/MFI)	Bank/MFI → External Fund	KCB Buni EFT or M-Pesa B2C	On lender approval	No (automated)	extloan.disbursed

Transaction Type	Direction	Rail	Meeting Step / Trigger	Member OTP Required?	Blockchain Event
IWL Direct Loan disbursement	IWL Lending Fund → Group	Paystack Transfer	On IWL Officer approval	No (automated to registered account)	iwl_loan.disbursed
IWL Direct Loan repayment	Group → IWL	Paybill or Paystack link	Scheduled repayment date	No (Paybill) / optional (Paystack)	iwl_loan.repayment_received
Paybill deposit (no bank acct groups)	Member/Treasurer → IWL Custodian	M-Pesa Paybill (522522)	Any time — not session-locked	No	paybill.received
Custodian fund allocation	IWL Custodian → Group Fund	Internal ledger transfer	Treasurer action, any time	No — Treasurer action in app	fund.credited
Share-out payout	Loan Fund → Members (batch)	M-Pesa B2C (batch)	Action Audit — Step 7 of Wizard	Yes (each member)	shareout.disbursed
Grant receipt	Donor/NGO → Grant Fund	EFT / KCB Buni	On partner approval — any time	No	grant.received
VSLF deposit	Group → Federation Pool	KCB Buni / M-Pesa	Post-meeting — on VSLF schedule	Treasurer OTP	payment.confirmed

10.2 IWL Paybill — Custodian Account

Attribute	Detail
M-Pesa Paybill Business Number	522522
Account Reference Format	GRP-XXXXX (e.g., GRP-00445) — system-generated, unique per group
Beneficiary	IWL Custodian Account — ring-fenced sub-ledger per group reference
Who May Deposit	Any M-Pesa subscriber — members, non-members, external parties on behalf of group

Attribute	Detail
Fund Allocation	Treasurer must allocate within 48h to: Loan Fund, Social Fund, External Loan Repayment, or IWL Loan Repayment
Unmatched Funds	Held in IWL unmatched holding account. IWL Finance investigates within 24h. Refunded via M-Pesa reversal if unresolvable.
Reversal Handling	M-Pesa reversal webhook handled by Paybill Service. Custodian balance debited. Treasurer notified. If already allocated: Fund Service reversal triggered.
Transition to Bank Account	When group opens a bank account: IWL Finance processes EFT of full custodian balance. Custodian sub-account closed. Future funds route directly to bank.
IWL Liability	IWL acts as custodian only. Funds held on trust for the group. IWL may not use custodian funds for its own operations.

10.3 Paystack Integration — IWL Direct Loans

Paystack API Endpoint	Purpose	IWL Business Rule
POST /transfer/initiate	Disburse IWL Direct Loan to group's registered M-Pesa or bank account	Only triggered after: (1) GA vote passed, (2) Credit Score ≥ 80 , (3) IWL Loan Officer manual approval on Web Platform, (4) Group has no active IWL loan
GET /transaction/verify/:ref	Confirm disbursement status	Passbook entry and blockchain record created only on 'success' status from Paystack
POST /refund	Reverse erroneous disbursement	Requires dual IWL Officer sign-off. Reversal window: 24 hours. Blockchain records reversal event.
Webhook: charge.success	Receive card or bank repayments from groups	Updates loan_repayments record; triggers Guarantee Service discharge check; credit score recalculated
Webhook: transfer.failed	Handle failed outbound disbursement	IWL Loan Officer notified immediately; loan status = 'disbursement_pending'; IWL initiates M-Pesa B2C fallback or re-initiates Paystack
POST /subaccount	Create segregated sub-account per group for fund management	Sub-account created at IWL loan approval. All IWL loan funds routed through group's Paystack sub-account.

10.4 External Integrations

Integration	Provider	Purpose	Fallback
Mobile Money (C2B/B2C)	Safaricom M-Pesa Daraja v3	STK Push, B2C payout, Paybill webhook, balance check, reversal, transaction status	Paystack (card payments) for select transactions; Cash Settlement Mode (see §13)
Payment Gateway	Paystack	IWL Direct Loan disbursements; card/bank repayments; sub-account management	M-Pesa B2C fallback for disbursements if Paystack unavailable
Banking API	KCB Buni API	Group bank accounts; bulk disbursement for external loans; account statements; loan application relay to KCB	EFT via SWIFT if Buni API unavailable; manual confirmation within 24h
National ID KYC	IPRS Kenya (Integrated Population Registration System)	Member National ID number + name verification at registration	Manual KYC hold: 'Pending Verification' status; document review within 48h by IWL staff
Credit Bureau	TransUnion CRB Africa	Member individual credit history check for external loan eligibility assessment	Skip check but flag in credit data package: 'CRB data unavailable at assessment'
SMS / USSD	Africa's Talking	OTP delivery, meeting reminders, arrears alerts, USSD fallback for feature phones	Emergency: IWL staff SMS via backup gateway (Twilio)
Market Prices	M-Farm Kenya	Live commodity prices for SPM+ business module members in agricultural VSLAs	Cached last-7-day prices served from BigQuery; prices labelled 'as of [date]'
Monitoring & Alerting	Grafana + Prometheus + Sentry	System health dashboards; API latency; payment success rates; error tracking; alerting	AWS CloudWatch as backup; PagerDuty for on-call escalation
Cloud Infrastructure	AWS (Africa — Kenya/Cape Town regions)	All compute, storage, networking, KMS key management, CI/CD	Multi-region within Africa; RTO 4h, RPO 1h

SECTION 11

UI Wireframes

Key screen layouts for developer reference — mobile-first (375px)

The wireframes below represent required screen layouts and component hierarchy. They are binding UX specifications, not suggestions. The design system uses IWL Green (#1B6B3A) as primary for all Member and Group Account screens, IWL Blue (#0D47A1) for Agent Accounts, and IWL Purple (#4A148C) for governance screens. All screens must support Kiswahili and English.

11.1 Onboarding — Account Type Selection

SECTION 2 — USER PERSONAS & ACCOUNT TYPES

Intelllcash Tri-Account Model

Three hard-bounded account types · Each determines every screen, permission, and data access path · Immutable after first registration

MEMBER

Individual VSLA member

"Save, borrow & vote in my group"

PERSONAS

Amina Wanjiku

Ordinary member · Own Android · 2G/3G · Low digital literacy

Joseph Mwangi

Chairperson · Group tablet · Cannot approve own loans

Grace Atieno

Treasurer · Manages B2C disbursements · Dual-OTP rules

Daniel Ochieng

Key-Holder · 6-digit OTP via SMS · GPS-verified

CORE ACCESS

Personal passbook — real-time view

Buy 1-5 shares via M-Pesa

Apply for internal loans

Anonymous GA votes

Guarantee fellow members' loans

Receive share-out payout

IWL Green (#1B6B3A) UI Theme

GROUP

VSLA group entity — managed by elected officers

"Manage our VSLA account collectively"

PERSONA

Unity Chama VSLA

Group-shared tablet managed by officers · Paybill accessible from any phone

CORE ACCESS

8-step meeting workflow management

Collect savings via M-Pesa Paybill

Apply for external loans (bank/MFI/IWL)

Conduct anonymous GA votes

Track all 4 fund sub-accounts

Build IWL Credit Score (0-100)

Share-Out Wizard at Action Audit

IWL Blue (#0D47A1) governance screens

AGENT

Village Agent / CBT · IWL Staff · Partner Officer

"Form, train, monitor & link groups"

PERSONAS

Peter Kamau (VA/CBT)

15-25 groups · IWL tablet · GPS field work · 3G/4G SIM

Margaret Waiganjo (IWL Staff)

IWL Loan Officer · Web Platform · Reviews IWL Direct Loans

Dr. Njoroge (Partner/MFI)

AMREF Partner Officer · Web browser · Read-only portfolio access

CORE ACCESS

CBT Module — Group Formation Wizard

GPS-stamped structured visit logs

9 IWL training module delivery

Phase advancement (requires Admin counter-sign)

VA Network (VAN) peer meetings

MEL dashboards & API (partners)

IWL Purple (#4A148C) Agent UI Theme

MOBILE APP — 5-TAB NAVIGATION BY ACCOUNT TYPE

TAB	MEMBER	GROUP (OFFICERS)	AGENT (CBT)
Dashboard	Personal shares, loan balance, guarantor liability, recent transactions, next meeting countdown	Group fund balances, credit score, active loans, meeting status, sync indicator	Caseload overview, overdue visits, pending phase advancements, performance
Meetings	View meetings, cast votes during session, confirm OTP for transactions	Start New Meeting, 3-key OTP activation, 8-step workflow, minutes recorder	Not primary tab — linked via CBT Module check-in gate
Members	View own profile, KYC status badge, passbook history	Full roster with search/filter, role assignment, member KYC management, credit scores	Group member roster for each caseload group
Main	Loan book: applications, repayment schedule, guarantor view	Full loan book with disbursement controls, guarantors, external finance	CBT: caseload dashboard, visit log, training modules, phase gate, VAN
More	Transactions, settings, account management, help & audio guides, sync status	Group Finance, Share-Out Wizard, Reports, settings, Paybill deposits	Group Formation Wizard, training modules, VAN network, performance reports

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11.2 Group Account — OTP Meeting Activation

SCREEN: ACTIVATE MEETING [Group Account → Meetings → Start Meeting]

☐ MEETING ACTIVATION

Unity Chama VSLA · Wednesday 7 May 2026 · 2:00 PM

Each Key-Holder must read their OTP aloud. Secretary enters it below.

Key-Holder	Name	Enter OTP	GPS Status	Result
Key-Holder 1 ☐	Daniel Ochieng	[_ _ _ _ _] → [VERIFY]	✓ 18m	☐ Waiting
Key-Holder 2 ☐	Akinyi Otieno	[4 8 2 9 1 7] → ✓	✓ 9m	✓ VERIFIED
Key-Holder 3 ☐	Mwangi Kamau	[_ _ _ _ _] → [VERIFY]	✗ 310m	✗ OUT OF RANGE

Meeting CANNOT start until all 3 OTPs are verified AND all 3 GPS checks pass.

11.3 Member Account — Personal Dashboard

SCREEN: MEMBER DASHBOARD [Member Account → Home]

My Shares (Total)

Loan Balance

Guarantor Liability

KES 14,400 (36 stamps)

KES 8,000 due 15 Jun 2026

KES 5,000 (Wanjiku's loan)

Recent Transactions

Date	Type	Amount	Status
07 May	Share Purchase ×2	KES 800	✓Confirmed
07 May	Social Fund	KES 100	✓Confirmed
30 Apr	Loan Repayment	KES 2,000	✓Confirmed

11.4 Group Account — External & IWL Loans Panel

SCREEN: GROUP FINANCE — EXTERNAL LOANS [Group Account → Group Finance → Loans]

☐ **IWL Credit Score: 84 / 100 · Status: Credit Eligible** ✓

Lender	Product	Max Amount	Rate	Term	Action
KCB Bank	VSLA Credit	KES 500,000	14% p.a.	12 mo	[APPLY]
Equity MFI	Group Loan	KES 300,000	18% p.a.	6 mo	[APPLY]
☐ IWL Direct	IWL VSLA Loan	KES 200,000	12% p.a.	9 mo	[APPLY]
AMREF MFI	Agri-Group	KES 150,000	16% p.a.	6 mo	[APPLY]

Applying triggers a GA vote (75% Yes required). Application cannot proceed without a successful vote.

11.5 Group Account — Paybill Savings Screen

SCREEN: PAYBILL SAVINGS [Group Account → Finance → Paybill Deposits]

☐ **Save via M-Pesa Paybill**

For groups without a bank account

Paybill Business Number	Your Group Account Reference	Group Name
522522	GRP-00445	Unity Chama VSLA

How to Pay (Jinsi ya Kulipa):

1. Open M-Pesa → Lipa Na M-Pesa → Pay Bill
2. Business No: 522522
3. Account No: GRP-00445
4. Enter amount and confirm with M-Pesa PIN

☐ Custodian Balance	☐ Unallocated Deposits	☐ Last Deposit
KES 47,200	KES 5,000 (needs allocation)	07 May 2026 — KES 2,000

11.6 Member Account — Credit Guarantee Request

SCREEN: LOAN GUARANTEE REQUEST [Member Account → ☐ Notification → Guarantee Request]

☐ **You Have Been Asked to Guarantee a Loan**

Borrower	Loan Amount	Purpose	Term	Your Liability if Default
Amina Wanjiku	KES 12,000	Secondary school fees	3 months	KES 12,000 (full — you are sole guarantor)

⚠ If Amina defaults on this loan, KES 12,000 will be deducted from your share balance at Share-Out. This commitment is recorded permanently on the blockchain.

Active Guarantees I Hold:

Borrower	Amount	Status	Due
Wanjiku Njoroge	KES 5,000	Active — repaying	June 2026

11.7 Agent Account — CBT Caseload Dashboard

SCREEN: AGENT DASHBOARD [Agent Account → CBT Module → Caseload]

☐ **Peter Kamau · Village Agent · Embu North Sub-County**
☐ Caseload: 14 Groups · 3 overdue visits · 1 pending phase advancement

Group	Phase	Last Visit	Quality	Next Visit Due	Action
Unity Chama	Intensive	07 May	87%	14 May	[LOG VISIT]
Imani VSLA	Development	22 Apr	91%	20 May	[LOG VISIT]
Tumaini Chama	Mobilisation	07 May	N/A	08 May	[FORMATION]
Umoja Group	Maturity	01 Apr	78% ⚠	01 May XOVERDUE	[VISIT NOW]

11.8 Web Platform — Partner Portfolio Dashboard

WEB SCREEN: PARTNER PORTFOLIO [Web Platform → Portfolio Overview]

AMREF Programme Portfolio · Embu & Kirinyaga Counties · Live Data

KPI	Value	vs. Last Month	Status
Total Groups	47	↑ +2	✓
Total Members	1,128	↑ +51	✓
Total Savings (KES)	3,842,000	↑ +12%	✓
Active External Loans	8 groups	— No change	✓
Average Repayment Rate	94.2%	↑ +1.3%	✓
Groups with Credit Score ≥75	31	↑ +3	✓
Groups Currently in Meeting	3	Live now	☐ Live

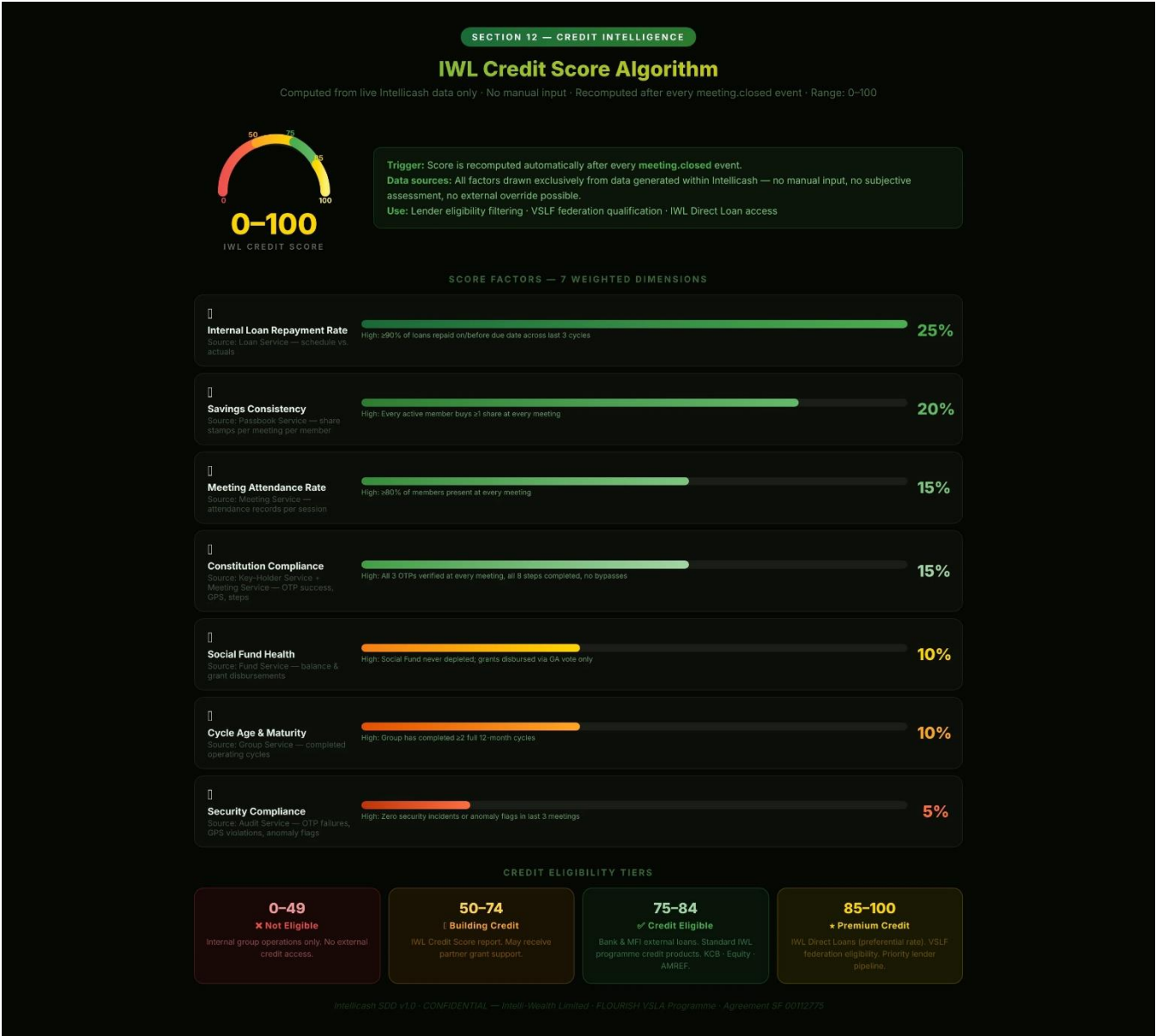
SECTION 12

IWL Credit Score & External Finance

Group creditworthiness model, external loan pipeline, and VSLF federation

12.1 IWL Credit Score Algorithm

The IWL Credit Score is a proprietary creditworthiness metric computed for every VSLA group after every meeting.closed event. It ranges from 0 to 100. The score is the primary gateway to external credit access and VSLF federation. It is computed entirely from data generated within Intellicash — no manual input, no subjective assessment.



Factor	Weight	Data Source	What a High Score Means
Internal Loan Repayment Rate	25%	Loan Service — repayment schedule vs. actuals	≥90% of loans repaid on or before due date in last 3 cycles
Savings Consistency	20%	Passbook Service — share stamps per meeting per member	Every active member buys at least 1 share every meeting
Meeting Attendance Rate	15%	Meeting Service — attendance records per session	≥80% of members present at every meeting
Constitution Compliance	15%	Key-Holder Service + Meeting Service — OTP success rate, GPS compliance, step completion	All 3 OTPs verified at every meeting; all 8 steps completed; no bypasses
Social Fund Health	10%	Fund Service — Social Fund balance; grant disbursements	Social Fund never depleted; grants disbursed by GA vote only
Cycle Age & Maturity	10%	Group Service — number of completed operating cycles	Group has completed ≥2 full 12-month cycles
Security Compliance	5%	Audit Service — OTP failures, GPS violations, anomaly flags	Zero security incidents or anomaly flags in last 3 meetings

Score Range	Credit Eligibility	Available Products
0–49	Not eligible	Internal group operations only; no external credit access
50–74	Building credit	Eligible for IWL Credit Score report; may receive partner grant support
75–84	Credit eligible	Bank and MFI external loan applications; standard IWL programme credit products
85–100	Premium credit	IWL Direct Loans at preferential rate; VSLF federation eligibility; priority lender pipeline

12.2 External Loan Pipeline (Web Platform)

The Credit Pipeline module on the Web Platform allows authorised banks and MFIs to view credit-eligible groups, filter by score, phase, county, and loan history, and access a one-click credit data package. This is a read-only, API-driven view — the platform never exposes group-level write actions to lenders.

Data in Credit Package	Source	Immutability
IWL Credit Score (current + 12-month trend)	Credit Score Service	Computed from blockchain-verified data; cannot be manually inflated
Meeting attendance history (last 12 meetings)	Meeting Service passbook	Blockchain-verified timestamps and GPS compliance
Savings trend (12-month aggregate)	Passbook Service	Append-only passbook; blockchain-backed
Internal loan repayment record (all cycles)	Loan Service	Append-only; defaulted loans visible — cannot be hidden
Constitution (current version with amendment history)	Group Service	SHA-256 hashed at each version; immutable
Group phase and cycle count	Group Service	Agent sign-off and Admin counter-sign required for phase changes
External loan repayment history	External Finance Service	Blockchain-backed; all defaults visible to all authorised lenders

12.3 VSLA Federation (VSLF)

VSLF Feature	Description	Eligibility
VSLF Membership	Groups join a federation pool to access larger collective credit tranches	Credit Score ≥ 85 ; ≥ 2 completed cycles; GA vote (75%); IWL Admin approval
Collective Loan Access	Federation applies as a collective to banks/MFIs; risk spread across member groups	All member groups must be Credit Score ≥ 75 at time of collective application
VSLF Deposit	Groups deposit a portion of their Loan Fund into the VSLF pooled account via KCB Buni	Minimum deposit: 10% of Loan Fund balance; maximum: 50%
Inter-Group Lending	Federation may on-lend to member groups from the pooled account at VSLF rates	Governed by VSLF constitution; requires VSLF committee vote; tracked in External Finance Service

SECTION 13

Risk Register & Contingency Plans

All known platform risks with mitigation strategies and fallback protocols

Every risk below has two responses: a Primary Mitigation (what the platform does to prevent or reduce the risk) and a Contingency Plan (the fallback protocol to be activated if mitigation fails or is insufficient). The contingency plans are binding operational procedures — they must be implemented in the platform and in IWL operational protocols. Risk ratings: Likelihood H=High M=Medium L=Low. Impact C=Critical H=High M=Medium.

Risk	Platform	Likelihood	Impact	Primary Mitigation	Contingency Plan (Fallback)
IWL methodology drift — platform features contradict IWLSGS	Both	M	C	Monthly methodology compliance audit; IWLSGS clauses encoded and locked in platform logic; cannot be disabled by any user role	Emergency feature flag disables offending function; IWLSGS review board convened within 48h; rollback deployed via forced app update to all devices
Low smartphone penetration — members cannot access Mobile App	Mobile	H	H	USSD channel (Phase 6); shared-device group model; CBT/VA acts as proxy device holder during meetings	Accelerate USSD delivery to Phase 3; negotiate device financing with Safaricom/KCB; paper-to-digital bridge mode retains paper passbooks alongside digital ledger
No mobile connectivity at meeting location — offline meeting needed	Mobile	H	H	IWL Sync Engine: full offline capability for all 8 steps; OTPs pre-cached; M-Pesa payment queue on reconnect	Manual Override Protocol: meeting recorded on paper; Secretary digitises within 24h; dual officer OTP signs off manual entry; anomaly logged to audit trail
Member distrust of digital system vs. physical cashbox	Mobile	H	M	CBT training programme; demo mode; gradual paper-to-digital transition; parallel paper + digital running period	Extend parallel period from 1 to 3 months; VA escalates persistent trust issues to IWL Admin; group may petition for extended hybrid mode via GA vote

Risk	Platform	Likelihood	Impact	Primary Mitigation	Contingency Plan (Fallback)
OTP SMS delay or non-delivery in rural areas	Mobile	H	H	USSD OTP fallback delivery via Africa's Talking; OTPs pre-cached when device has signal	If both SMS and USSD fail: IWL Admin activates Manual Override Protocol with mandatory audit note; session_id created manually; blockchain entry includes override flag
Three Key-Holder collusion to open meeting fraudulently	Mobile	L	H	All-member real-time ledger visibility during meeting; GPS independent verification; anomaly detection in Audit Service; partner dashboard alerts	Automatic meeting freeze on anomaly detection; IWL Admin and partner alerted via Web Platform; forensic audit of session logs; group placed on Enhanced Monitoring; key-holders replaced via emergency GA vote
M-Pesa API downtime during active meeting	Mobile	M	H	Payment queue + retry engine; meeting continues with manual recording in app; auto-reconciliation on reconnect	If downtime >2h: Treasurer activates Cash Settlement Mode — transactions recorded digitally with 'Pending Payment' status; members acknowledge in-app; all payments processed within 24h on reconnect; reconciliation report generated and partner-visible
External loan group default cascading to members	Both	M	H	Conservative IWL credit threshold (≥ 75 required); lender holds credit risk, not IWL; external fund completely segregated from internal fund	Default Early Warning System: repayment rate <70% triggers auto-alert to IWL Admin, partner, and VA; escalation visit within 5 days; loan restructuring workflow offered; if default confirmed, share-out suspended; legal protocol per IWL constitution

Risk	Platform	Likelihood	Impact	Primary Mitigation	Contingency Plan (Fallback)
IWL Direct Loan default — Paystack disbursement unrecoverable	Both	M	H	Credit score ≥80 gate; GA vote required; IWL Officer manual review; Guarantee Service calls guarantees on default	Arrears flag at Day 7; credit score penalised from Day 1; IWL Loan Officer initiates restructuring; legal debt recovery per IWL Loan Agreement if restructuring fails; write-off requires IWL Board approval and audit log entry
Paybill unmatched deposits — wrong account reference	Both	H	M	Clear paybill reference format (GRP-XXXXX) displayed prominently; confirmation SMS with group name	Unmatched funds held in IWL holding account; IWL Finance investigates within 24h; refunded via M-Pesa reversal if unresolvable within 72h; all unmatched events reported to IWL CFO weekly
Guarantor cannot cover called guarantee at share-out	Mobile	M	H	Guarantor passbook flagged immediately on acceptance; guarantor cannot receive own share-out until guarantee resolved	Guarantor's share-out held in escrow; borrower given 30-day cure period; if uncured: guarantor's escrow applied; any residual written off with GA vote and documented in audit log; credit score impact for both borrower and guarantor
Vote manipulation via social pressure despite anonymity	Mobile	L	M	Ballot encryption enforced; no officer can view individual votes; result hashed immediately	If result disputed: any member can raise formal objection in-app within 24h; anonymised audit (counts only) reviewed by IWL Admin; if irregularity confirmed, re-vote within 7 days facilitated by VA
CBK regulatory non-compliance (PSSP licensing)	Both	M	C	Early CBK engagement; sandbox licence in process; pass-through model (IWL never holds funds as financial institution); CBK-aligned transaction reporting	Platform operates in restricted mode: no direct M-Pesa B2C; all disbursements via KCB Buni as licensed intermediary; operations paused for non-compliant types until licence confirmed; compliance counsel engaged

Risk	Platform	Likelihood	Impact	Primary Mitigation	Contingency Plan (Fallback)
Blockchain node failure or compromise	Both	L	C	3-node Raft consensus with 2-of-3 endorsement; geographic node separation; bi-annual penetration testing	If 1 node compromised: 2-of-3 endorsement maintains operation; compromised node isolated; IWL security team forensic review within 4h; node rebuilt from backup; all transactions validated against remaining nodes; CBK notified if data integrity at risk
Member personal data breach	Both	L	C	AES-256 encryption; bi-annual penetration testing; KDPA compliance; minimal PII collection; IWL DPO oversight	Breach Response Plan: system isolation within 1h; IWL DPO notified immediately; ODPC notified within 72h; affected members notified via SMS; forensic audit within 48h; partner notification via Web Platform; affected credentials reset; public disclosure per KDPA Article 43 if threshold breached
Sync conflict corrupting group ledger after offline meeting	Mobile	L	C	CRDTs ensure conflict-free merge; append-only ledger; server always authoritative for blockchain-signed entries	If conflict detected: affected meeting record flagged 'Conflict Detected'; IWL Admin alerted on Web Platform; conflicting entries presented to Treasurer + Chairperson for manual adjudication within 48h; dual officer OTP required to resolve; passbooks locked read-only until resolved
VA digital literacy gap — agent cannot operate CBT Module	Mobile	H	M	CBT App icon-first design; audio prompts; Kiswahili; IWL VA digital training before field deployment	If VA fails digital literacy threshold: 2-week hands-on coaching by IWL Master Trainer; VA shadowing experienced VA for first 3 groups; minimum score gate enforced before VA can advance groups to Intensive Phase

Risk	Platform	Likelihood	Impact	Primary Mitigation	Contingency Plan (Fallback)
Key third-party API failure (M-Pesa / IPRS / KCB)	Mobile	M	H	Redundant API providers; Paystack as M-Pesa fallback for card payments; retry logic with exponential backoff	M-Pesa outage → Paystack; IPRS outage → manual KYC hold ('Pending Verification', full access within 48h); KCB outage → defer external disbursements; all fallback states logged and partner-visible on Web Platform status page
IWL Custodian account misused by IWL for own purposes	Both	L	C	Fund ring-fencing enforced at DB and service level; custodian balance tracked separately; monthly independent audit; partner-visible on Web Platform	If misuse suspected: immediate forensic audit by IWL Board; external auditor engaged; custodian balance verified against M-Pesa transaction history; affected groups notified; regulatory report to CBK if confirmed

SECTION 14

REST API Specification

OpenAPI 3.0 endpoint reference for partners, banks, and integrators

Intellicash exposes a REST API v1 (OpenAPI 3.0) documented at api.intellicash.co.ke. All partner access is via OAuth2 client credentials scoped to their programme groups. All endpoints are read-only for external partners. Write access is reserved for the Mobile App and Web Platform. IWL owns and controls the specification.

14.1 Groups & Portfolio Endpoints

Method	Endpoint	Description	Auth / Scope
GET	/v1/groups	List all VSLAs in partner scope; filter by phase, credit_score, county, GPS compliance, VSLF, external_loan_status	OAuth2 — partner scope
GET	/v1/groups/{id}	Full group profile: constitution, member count, phase, GPS, VA, IWL credit score, cycle	OAuth2 — partner scope
GET	/v1/groups/{id}/meetings	Meeting history: dates, attendance, 3-key status, step completion, transaction totals	OAuth2 — partner scope
GET	/v1/groups/{id}/ledger	Aggregate ledger: savings, loan book, Social Fund, External Fund, repayment rate	OAuth2 — partner scope
GET	/v1/groups/{id}/passbook/{memberId}	Individual passbook: stamps, loans, Social Fund history, share-out (scoped — partner must have member consent)	OAuth2 — partner scope + member consent
GET	/v1/groups/{id}/credit-score	IWL Credit Score (0–100), factor breakdown, 6-month trend, eligibility flags	OAuth2 — lender scope
GET	/v1/groups/{id}/external-finance	External loans, IWL direct loans, grants, VSLF status, repayment records	OAuth2 — lender scope
GET	/v1/groups/{id}/votes	All resolutions: motion, type, result, quorum compliance, blockchain hash, date	OAuth2 — partner scope

Method	Endpoint	Description	Auth / Scope
GET	/v1/portfolio	Aggregate portfolio KPIs: total savings, members, repayment rate, phase distribution, security compliance for partner scope	OAuth2 — partner scope

14.2 MEL & Reporting Endpoints

Method	Endpoint	Description	Auth / Scope
GET	/v1/analytics/mel-report	MEL export with 47 indicators; filterable by date range, county, phase	OAuth2 — partner scope
GET	/v1/analytics/cbts	VA performance: visit compliance, quality scores, group phase distribution, VAN activity	OAuth2 — partner scope / IWL Admin
GET	/v1/analytics/credit-pipeline	Groups eligible for external lending; filterable by score threshold, phase, county	OAuth2 — lender scope
GET	/v1/audit/events	Security audit log: OTP failures, GPS violations, anomaly flags (IWL Admin only)	OAuth2 — IWL Admin scope
POST	/v1/webhooks/subscribe	Register webhook URL for VSLA lifecycle event push notifications	OAuth2 — partner scope
DELETE	/v1/webhooks/{id}	Unsubscribe a webhook endpoint	OAuth2 — partner scope

14.3 Webhook Events

Event	Trigger	Payload Fields	Partner Use Case
group.phase_advanced	Group lifecycle phase changes	group_id, old_phase, new_phase, timestamp	NGO monitoring; programme milestone reporting
meeting.closed	Meeting Step 8 completed	group_id, session_id, attendance_count, total_savings, total_loans, timestamp	Real-time portfolio dashboard updates

Event	Trigger	Payload Fields	Partner Use Case
loan.disbursed	Internal loan M-Pesa B2C confirmed	group_id, loan_id, amount, vote_hash, timestamp	Partner loan portfolio monitoring
extloan.disbursed	External loan or IWL loan disbursed	group_id, application_id, lender_type, amount, timestamp	Bank/MFI loan confirmation; IWL Finance alert
credit_score.threshold_crossed	Credit Score crosses 75 or 85	group_id, old_score, new_score, threshold_crossed, timestamp	Bank/MFI eligibility notifications
loan.defaulted	Loan marked as defaulted by arrears engine	group_id, loan_id, principal_outstanding, guarantors_count, timestamp	Partner early warning; lender alert
share_out.completed	Action Audit Share-Out finalised	group_id, cycle_number, total_disbursed_kes, member_count, timestamp	Programme cycle completion reporting
group.created	New VSLA group activated by IWL Admin	group_id, county, phase, agent_id, timestamp	Portfolio growth tracking; MEL baselines

SECTION 15

Implementation Roadmap

Phased build plan — Month 0 to Month 30

This roadmap is binding. All features listed in a phase must be delivered and tested before the next phase begins. No phase may be skipped. IWL Executive Director must formally sign off each phase before development on the subsequent phase commences.

Phase	Scope	Key Milestone	Dependencies
Phase 0 Foundation	IWLSGS v1.0 finalisation; tri-account onboarding architecture sign-off; Hyperledger Fabric network setup (3 nodes); PostgreSQL schema design; API Gateway configuration; AWS Kenya Region provisioning; GitHub Actions CI/CD pipeline	Architecture, spec, and infrastructure signed off by IWL Executive Director	None — this is the foundation
Phase 1 Core Group & Identity	Identity Service; Group Service; Group Formation Wizard (Agent Account); Member Registration + IPRS KYC; IWL Sync Engine (offline SQLite + CRDT merge); CBT Module — Formation phase; Blockchain Service (meeting_cc + passbook_cc chaincodes)	First IWL VSLA group registered digitally; first member passbook created	Phase 0 infrastructure live; IPRS API access confirmed; Africa's Talking SMS
Phase 2 Meeting Workflow	Key-Holder Service + Three-Key OTP Protocol; 8-Step Meeting Workflow Engine; Digital Passbook; M-Pesa STK Push (basic C2B for shares, social fund, fines); Meeting Service; Passbook Service (append-only, blockchain-backed); OTP-Activated Meeting fully live	First fully digital IWL meeting conducted with Three-Key protocol	Phase 1 complete; M-Pesa Daraja v3 sandbox access; Africa's Talking USSD fallback
Phase 3 Loans & Governance	Loan Service + internal loan workflow; Credit Guarantee Service; Vote Service (anonymous ballot + blockchain hash); Social Fund (full); M-Pesa B2C disbursements; Fund Service (full 3-namespace); Share-Out Wizard v1; Partner Web Platform v1 (read-only portfolio); Blockchain Service (loan_cc + vote_cc chaincodes)	First internal loan disbursed via GA vote; first anonymous ballot; partners view live data on Web Platform	Phase 2 complete; M-Pesa Daraja v3 production access; KCB Buni API sandbox
Phase 3.5 Paybill Custodian	Paybill Service; M-Pesa Paybill C2B webhook; IWL Custodian Account sub-ledger; Group Account Paybill savings screen; Treasurer allocation workflow; Blockchain (payment_cc chaincode for paybill events)	First group deposit via Paybill; first Treasurer allocation to fund	M-Pesa Paybill business number 522522 registered; Phase 3 Fund Service live

Phase	Scope	Key Milestone	Dependencies
Phase 4 External Finance	External Finance Service; External Loan Module (Bank/MFI pipeline); IWL Credit Score Service (full 7-factor algorithm); IWL Direct Loan Service (Paystack integration); Paystack Loan Service; Credit Pipeline on Web Platform; KCB Buni external loan integration; Blockchain (extloan_cc chaincode); VSLF federation module	First external loan facilitated; first IWL Direct Loan disbursed via Paystack; credit score visible to lenders on Web Platform	Phase 3 complete; Paystack API production access; KCB Buni API production; Credit Score model trained on Phase 1–3 data
Phase 5 API & Advanced Analytics	REST API v1 (OpenAPI 3.0); Webhook system (all events); API Developer Portal on Web Platform; Analytics Service (47 MEL indicators); MEL report exports (PDF/Excel); SPM+ business modules; IWL Admin system configuration modules; Partner API onboarding	API live with first partner MIS integration; all 47 MEL indicators reporting	Phase 4 complete; BigQuery data warehouse provisioned with Phase 1–4 data
Phase 6 USSD & Inclusion	USSD channel for feature phones (Africa's Talking); Microinsurance linkage; Gender sub-wallet reporting; M-Farm market price feeds; Accessibility improvements (enhanced audio prompts); Multi-language expansion (Dholuo, Kikuyu)	Feature phone users can participate in VSLA meetings via USSD; first microinsurance product offered	Phase 5 complete; Africa's Talking USSD production integration
Phase 7 East Africa Expansion	Uganda, Tanzania, Ethiopia regulatory compliance; Multi-language (Luganda, Swahili variants, Amharic); Cross-border VSLF lending; Multi-currency ledger; Country-specific KYC (NIRA Uganda, NIDA Tanzania)	First non-Kenya VSLA group registered; East Africa platform operational	Phase 6 complete; regulatory approval in target countries; partnerships with in-country MNOs

SECTION 16

Web Platform Specifications

Module map, access control, and feature requirements

16.1 Web Platform Module Map

Module	Available To	Core Features	Data Sources
Portfolio Overview	Partner Officers, IWL Admin	Aggregate KPIs for all programme groups; groups by phase; total savings trend; repayment rate; OTP/GPS compliance rate; live meeting count	Analytics Service; Meeting Service; Credit Score Service
Group Intelligence	Partner Officers, IWL Admin	Per-group scorecard: phase, meeting cadence, savings trend, loan book performance, IWL Credit Score history (12-month trend); officer roster	Group Service; Credit Score Service; Passbook Service
Live Meeting Monitor	Partner Officers, IWL Admin	Groups currently in session; 3-key unlock status; GPS compliance; running transaction totals; step progress; real-time alerts for anomalies	Meeting Service; Key-Holder Service; Audit Service
Credit Pipeline	Banks, MFIs, IWL Admin	Groups eligible for external lending (credit score filter); per-group credit data package; one-click loan application intake; active loan tracker	Credit Score Service; External Finance Service
IWL Loan Management	IWL Loan Officers, IWL Admin	IWL Direct Loan application queue; officer review and approval; Paystack disbursement initiation; repayment monitoring; arrears management	Paystack Loan Service; External Finance Service
Grant & External Loan Management	Partner Officers, IWL Admin	All external loan and grant applications by status; approval workflow; disbursement tracking; repayment schedules; partner reporting	External Finance Service; Fund Service
CBT Performance	Partner Officers, IWL Admin	VA visit compliance; quality score trends; groups at risk; VA caseload distribution; VAN activity; phase advancement log	CBT Service; Group Service
MEL & Impact Reports	Partner Officers, IWL Admin	47 IWL standard MEL indicators; customisable indicator sets; automated PDF/Excel/CSV export; API feed; historical trend charts	Analytics Service; BigQuery

Module	Available To	Core Features	Data Sources
Security & Audit Log	IWL Admin only	OTP failure log; GPS out-of-range alerts; constitution deviation flags; failed transaction attempts; AML flags; immutable audit trail search	Audit Service; Blockchain Explorer
Votes & Governance	Partner Officers, IWL Admin	All resolutions: type, motion, result, quorum compliance, blockchain hash; officer election history; constitution version history	Vote Service; Group Service
Partner Administration	IWL Admin only	Partner onboarding; OAuth2 API key management; data scope configuration; webhook subscriptions; sandbox environment management	Identity Service; API Gateway config
VA & CBT Onboarding	IWL Admin only	VA registration; caseload assignment; digital literacy assessment scores; training records; performance dashboard	CBT Service; Identity Service
System Configuration	IWL Admin only	IWLSGS version management; constitution templates; service charge defaults; GPS radius defaults; regulatory report generation; feature flags	All services — config namespace
API Developer Portal	Partner Officers, Developers	OpenAPI 3.0 interactive documentation; sandbox environment; webhook testing console; OAuth2 client management; rate limit monitoring	API Gateway; Analytics Service

SECTION 17

Glossary

Definitive terminology for Intellicash v1.0

Term	IWL Definition
Action Audit	IWL's end-of-cycle full audit and share-out ceremony; digitised as the Share-Out Wizard in the Mobile App
Agent Account	Intellicash account type for Village Agents / CBTs and IWL staff. Provides CBT Module, caseload management, group formation, and (for IWL staff) loan officer functions.
Blockchain Service	Intellicash microservice wrapping Hyperledger Fabric. Receives async financial events via Kafka; commits immutable transaction hashes; returns confirmation hashes to originating services.
CBT	Community-Based Trainer — IWL field facilitator who forms and trains VSLAs; also called Village Agent (VA).
Credit Guarantee	Formal commitment by one or more VSLA members to repay a fellow member's loan if that member defaults. Recorded on blockchain; reflected on guarantor passbook as contingent liability.
Credit Score Service	Microservice computing IWL Credit Score after every meeting.closed event using 7 weighted factors from live group data.
Custodian Account	IWL-managed M-Pesa Paybill sub-account holding VSLA savings for groups without bank accounts. Funds are ring-fenced per group reference; IWL acts as custodian, not beneficial owner.
External Loan Sub-Account	Segregated fund namespace within the Group Account for bank/MFI external loan receipts. Never commingles with Internal Loan Fund or Social Fund.
Fund Service	Microservice managing three group fund namespaces: loan_fund, social_fund, external_fund. No cross-namespace transfer without explicit Treasurer action + dual officer OTP.
General Assembly (GA)	Full membership of a VSLA; the supreme governing body. All major decisions require a GA vote conducted during an active Three-Key unlocked meeting session.
Grant Fund	Segregated fund namespace for non-repayable grant receipts from NGO/donor/government partners. Purpose-locked; every disbursement requires GA vote; partner-visible on Web Platform.
Group Account	Intellicash account type representing a VSLA group entity. Used by officers and key-holders to manage meetings, funds, external loans, Paybill savings, and GA votes.
Guarantee Service	Microservice managing the lifecycle of credit guarantees: nomination, acceptance, active liability tracking, calling on default, and discharge on full repayment.

Term	IWL Definition
Hyperledger Fabric	Enterprise-grade private permissioned blockchain used by Intellicash for immutable financial event records. Separate channel per group; off-chain data storage with on-chain hash verification.
IWL Credit Score	Proprietary IWL creditworthiness algorithm (0–100) computed after every meeting. Used to determine eligibility for external loans, IWL Direct Loans, and VSLF federation.
IWL Direct Loan	A loan product offered directly by Intelli-Wealth Limited to qualifying VSLA groups (Credit Score ≥ 80). Disbursed via Paystack Transfer; governed by mandatory GA vote; repaid via Paybill or Paystack link.
IWL Sync Engine	IWL proprietary offline-first synchronisation layer. Uses CRDTs for conflict-free merge on reconnect. Enables full 8-step meeting operation without internet connectivity.
IWLSGS	IWL Savings Group Standard — IWL's proprietary VSLA methodology specification. Every Intellicash feature must map to an IWLSGS procedure.
Key-Holder	One of three designated VSLA members who each receive one OTP for meeting activation. Key-Holders cannot simultaneously hold MC officer roles (system-enforced).
Loan Fund	Internal fund namespace accumulating member share purchases, fines, and interest. Only disbursed as internal member loans; governed by 3× savings cap; separate from Social Fund.
Management Committee (MC)	Five elected group officers: Chairperson, Secretary, Treasurer, Money Counter 1, Money Counter 2. None may serve as Key-Holder.
Member Account	Intellicash account type for individual VSLA members. Provides passbook view, share purchase, loan application, anonymous voting, and loan guarantee functions.
OTP	One-Time Password — 6-digit code generated by CSPRNG; delivered via SMS or USSD; used for meeting Three-Key activation and individual transaction confirmation.
Operating Cycle	One full 12-month VSLA year, ending at Action Audit and Share-Out. Group cycle counter increments on Share-Out completion.
Passbook Service	Microservice maintaining each member's append-only personal ledger: share stamps, loan records, Social Fund history, and Share-Out archive. Backed by blockchain.
Paybill Service	Microservice processing M-Pesa Paybill (Business No. 522522) C2B webhooks; matching account references to groups; crediting custodian sub-accounts; alerting Treasurers.
Paystack Loan Service	Microservice managing IWL Direct Loan lifecycle: group application, IWL officer review queue, Paystack disbursement API, and repayment reconciliation.
Resolution	A motion put to a GA vote during an active meeting session. Types include loans, grants, elections, constitution changes, expulsions, and external loan approvals.

Term	IWL Definition
Service Charge	IWL term for loan interest; expressed as flat % per month × loan term (not compound). Set in group constitution.
Share	Unit of member savings; member buys 1–5 shares per meeting. Loan ceiling = 3× share value. Share value is the share denomination set in group constitution.
Share-Out	Distribution of accumulated savings plus returns to members at Action Audit. Disbursed via M-Pesa B2C batch. Requires GA vote to approve schedule.
Social Fund	Emergency grant pool; fixed member contribution per meeting; grants only (never loans); GA vote required for every disbursement.
Three-Key Protocol	The digital equivalent of IWL's physical three-padlock cashbox system. All 3 Key-Holders must verify via OTP and GPS before any meeting session can open.
Tri-Account Onboarding	The first-launch screen where every new user selects one of three account types: Member, Group, or Agent. This choice is permanent and determines all UI paths and permissions.
VAN	Village Agent Network — peer network of IWL VAs for post-project sustainability, mutual support, and cross-group learning.
VSLF	VSLA Federation — IWL wholesale credit model: groups pool capital collectively to access external credit at better terms.
Web Platform	Browser-based Intellicash application (Next.js) serving Partner Officers, Banks/MFIs, and IWL Administrators with portfolio intelligence and programme management tools.

AUTHORISED RELEASE

Intellicash System Design Document — Version 1.0

This document supersedes all prior versions, wireframes, and partial specifications.

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All amendments to this document require Executive Director sign-off and version increment.